

Defining and applying an electricity demand flexibility benchmarking metrics framework for grid-interactive efficient commercial buildings

Jingjing Liu^{a,*}, Rongxin Yin^a, Lili Yu^a, Mary Ann Piette^a, Marco Pritoni^a, Armando Casillas^a, Jiarong Xie^b, Tianzhen Hong^a, Monica Neukomm^c, Peter Schwartz^d

^a E O Lawrence Berkeley National Laboratory, Berkeley, CA, United States

^b Carnegie Mellon University, Pittsburgh, PA, United States

^c U.S. Department of Energy, Washington DC, United States

^d former employee of E O Lawrence Berkeley National Laboratory

ARTICLE INFO

Keywords:

Demand flexibility

Metrics

Benchmarking

Demand response

Field data

Global Temperature Adjustment (GTA)

ABSTRACT

Building demand flexibility (DF) research has recently gained attention. To unlock building DF as a predictable grid resource, we must establish a quantitative understanding of the resource size, performance variability, and predictability based on large empirical datasets. Researchers have proposed various sets of theoretical metrics to measure this performance. Some metrics have been applied to simulation results, but most fall short of exploring the complexities in real building applications. There are practical metrics used in individual demand response field studies but they alone cannot fulfil the job of DF benchmarking across a diverse group of buildings. The electrical grid's geographically diverse and changing nature presents challenges to comparing building DF performance measured under different conditions (i.e., *benchmarking DF*). To address this challenge, a novel DF benchmarking framework focused on load shedding and shifting is presented; the foundation is a set of simple, proven single-event metrics with attributes describing event conditions. These enable benchmarking and visualization in different dimensions for identifying trends that represent how these attributes influence DF. To test its feasibility and scalability, the DF framework was applied to two case studies of 11 office buildings and 121 big-box retail buildings with demand response participation data. These examples provided a pathway for using both building level benchmarking and aggregation to extract insights into building DF about magnitude, consistency, and influential factors. Potential applications of the framework and real-world values have been identified for grid and building stakeholders.

1. Introduction

1.1. Demand flexibility research background

Building Demand Flexibility (DF) related research studies have become a popular topic in recent years. They are driven by the challenge of balancing electric supply and demand presented by the increasing penetration of renewable energy generation sources on the grid (that increase the dynamics and uncertainties of electricity supply) and the emergence of distribution energy resources (DERs) (that increase the dynamics and uncertainties of electricity demand). A couple of major initiatives have propelled the research of DF internationally and in the United States. The International Energy Agency's (IEA) Energy in Buildings and Communities Programmes (EBC) has been organizing research collaboration on this topic for several years, including Annex 67 [1] "En-

ergy Flexibility in Buildings" and Annex 82 [2] "Energy Flexible Buildings Towards Resilient Low Carbon Energy Systems," which is a follow-up of Annex 67. Annex 67 defined "energy flexibility (EF)" as a building's ability to manage its demand and generation according to local climate conditions, user needs, and energy network requirements [3]. It also introduced a methodology to characterize EF by quantifying the amount of energy a building can shift according to an external forcing factor ("penalty signal" from the surrounding energy networks) while maintaining occupant comfort [4]. Given its complexity and academic nature, the methodology is to be further evaluated with real building data for its value to the grid and building stakeholders. IEA EBC Annex 81 [5], Data-driven Smart Buildings, has an activity under subtask C to study data-driven DF indicators and quantification methods. The activity also compiles a list of building performance datasets that can be used to test and benchmark methods for quantifying DF of buildings.

Abbreviations: DF, demand flexibility; DR, demand response; LF, load flexibility; EF, energy flexibility; GTA, global temperature adjustment.

* Corresponding author at: E O Lawrence Berkeley National Laboratory, 1 Cyclotron Rd, Mail Stop: 90R3147, Berkeley, California 94720, United States

E-mail address: JingjingLiu@lbl.gov (J. Liu).

<https://doi.org/10.1016/j.adapen.2022.100107>

Received 27 July 2022; Received in revised form 31 August 2022; Accepted 15 September 2022

Available online 17 September 2022

2666-7924/© 2022 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license

(<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Nomenclature

CV (RMSE)	Coefficient of Variation of the Root-Mean-Square Error
DDI	Demand Decrease Intensity
DDP	Demand Decrease Percentage
DDSD	Demand Decrease Intensity Standard Deviation
DERs	Distribution Energy Resources
DF	Demand Flexibility
DII	Demand Increase Intensity
DIP	Demand Increase Percentage
DISD	Demand Decrease Intensity Standard Deviation
DOE	Department of Energy
DR	Demand Response
EBC	Energy in Buildings and Communities
EE	Energy Efficiency
EF	Energy Flexibility
EUI	Energy Use Intensity
GEB	Grid-Interactive Efficient Buildings
HVAC	Heating, Ventilation, and Air Conditioning
IEA	International Energy Agency
KPI	Key Performance Indicators
LF	Load Flexibility
nMBE	Normalized Mean Bias Error
OAT	Outside Air Temperature
RTU	Rooftop Units
SPDI	Summer Peak Demand Intensity
VAV	Variable Air Volume
WPD	Winter Peak Demand Intensity

In 2019, the United States Department of Energy (DOE) launched a Grid-Interactive Efficient Buildings (GEB)¹ initiative that aims to optimize DERs for the benefit of building owners, occupants, and the electric grid [6]. DOE has published a series of technical reports evaluating the technological capabilities and potential of residential and commercial buildings for grid services; the report series covers major relevant building technology areas such as heating, ventilation and air conditioning (HVAC); lighting; envelope; and controls [7]. The GEB initiative defines “demand flexibility” as “the capability of DERs to adjust a building’s load profile across different timescales; energy flexibility and load flexibility (LF) are often used interchangeably with demand flexibility” [7]. The GEB initiative also identifies four demand-side management (DSM) strategies covered under DF (aside from energy efficiency, EE) – load shedding (*Shed*), load shifting (*Shift*), modulating, and generation – amongst which, *Shed* and *Shift* are prioritized. Each of these DSM strategies are mapped to a range of grid services² such as contingency reserves, transmission and distribution capacity service, and renewable curtailment mitigation [7]. *Shed* and *Shift* are also focuses of the research presented in this paper. Demand flexibility resources from buildings are often aggregated before offered as a single resource to the grid. Such aggregation can not only help meeting the minimum resource size thresholds but also stagger individual building loads in time to achieve a desirable overall load change profile. Therefore, the benchmarking metrics presented in this paper are also useful to aggregators.

DF from *Shed* and *Shift* can be triggered by grid despatch or time-varying electricity prices such as critical peak pricing and real-time pricing.

One important premise of DF is to shift load from high greenhouse gas emission (i.e., “dirty”) periods to the renewable energy abundant (i.e., “clean”) periods to achieve decarbonization. Shedding load also has a unique role in maintaining grid reliability. In fact, for decades grid operators have been using *Shed* as an important lever when the grid is under peak or emergency conditions. This is also known as *demand response* (DR). Therefore, there is a strong connection between DF (or EF, LF) and DR, although there are differences between the two. A group of GEB researchers articulates that “like DR, DF is characterized by active load management on timescales consistent with utility system and grid needs. Unlike EE and DR, DF is not a resource in the traditional sense, but a potential that the utility or system operator can utilize to provide reliable electricity service. From the system operator’s perspective, EE and DR are what you have in your portfolio and DF is what you can do with the resources you have [8].” Therefore, studies of DR practices and performances are relevant in the DF metrics literature review in the following sections.

1.2. The need for demand flexibility benchmarking

The authors believe that building DF benchmarking is primarily needed to benefit planning and post-analysis activities, which in turn informs operation, for both grid entities and building stakeholders. This section explains the motivation of this paper.

With building DF gains attention as an emerging resource for grid decarbonization, there has been noteworthy research work that aims at quantifying the DF resource size at a national or state level based on either prototype whole-building energy simulation combined with building stock data [9] or estimated load change by end use applied on cluster load shapes derived from samples [10]. There has been little understanding of how the DF performance of real buildings compares with the technical potential estimated based on building energy simulation to support applications of these estimates in grid planning. Without such understanding, grid operators and utilities cannot estimate building DF potential with confidence. Nor can they verify the hypothesis created from simulations regarding which building types, climates, vintages and other factors would lead to greater DF for program design and other planning considerations. To understand the actual DF resource size from the building stock and what contributes to greater DF for a building, we must review field-measured DF performance from a wide range of buildings and compare them to support more robust bottom-up estimates. This invites the questions of how to measure DF performance using metrics and how to connect them with potential influential factors. The answer can be complex because building DF is dependant on time (day of week, hour of day, and season) and other factors such as deployed strategy and weather.

Another significant concern of flexible building load resources in grid planning is about their performance consistency and predictability. When building DF is used beyond a simple economic resource and for grid reliability, the unknown consistency of field performance presents a significant challenge to grid planning and operations. Large datasets of actual performance are critical to establishing basic understanding of performance variations. Predictability can then be better understood based on empirical correlation studies between DF performance and various independent influential factors when such data are available and added to the whole building power data. Currently, there are a number of DR programs in the U.S., but there is a lack of publicly available DR performance data, along with building information, that can support such research. The frequency, duration, and timing of DR events can vary significantly amongst geographic locations and utilities. Furthermore, for the same location and utility, the timing of the DR events also evolves over time as renewables and electrification grow. Therefore, performance data collected in the field may not be directly comparable if they are associated with different timing (e.g., afternoon versus evening) and other factors (e.g., hot day versus mild day). The framework used to collect DF performance data and compare metrics across

¹ A GEB is defined as “An energy-efficient building that uses smart technologies and on-site DERs to provide demand flexibility while co-optimizing for energy cost, grid services, and occupant needs and preferences, in a continuous and integrated way. [7]”

² “Grid services refer to services that support the generation, transmission, and distribution of electricity and provide value through avoided electricity system costs (generation and/or delivery costs). [7]”

buildings must address this issue; otherwise, it would lead to misuse of data.

From the building managers' perspective, the above issues of performance consistency and predictability are also relevant. Without the knowledge of how performance can change under the influence of various influential factors, building managers are subject to under-performance penalties, lost revenues from overly conservative estimates, and other adverse effects. In addition, **comparing the DF performance of a peer group of similar buildings can help identify both best practices to be promoted and performance issues to be diagnosed**. This is also what we refer to as **"DF benchmarking"**.

In terms of selecting a cohort of buildings that are similar, we expect that the necessary data source will grow over time for DF benchmarking as it did for building energy benchmarking. The latter uses *energy use intensity*, or EUI (kBtu/ft²/yr, or GJ/m²/yr) as the metric to compare buildings' annual energy use. Multiple large energy benchmarking data sources and tools have been established, such as ENERGY STAR's Portfolio Manager [11] and Building Performance Database [12], to enable building managers and practitioners to compare the efficiency of their buildings of interest with similar buildings. As these benchmarking databases grew larger over time, it became possible to benchmark a specific building against a cohort with shared characteristics, such as size, vintage, space use, operating hours, HVAC equipment type, and others.

1.3. Demand flexibility metrics in the literature

To benchmark building DF performance, we need consistent metrics. Our literature review of Shed and Shift related DF metrics revealed that the existing DF metrics definitions and framework focus on prototype-based building energy simulation results and have not been applied to the actual performance of a significant group of real buildings to test their feasibility. Below we review representative existing metrics and discuss the reasons why we selected some of them for our proposed benchmarking framework and why others were not selected.

1.3.1. Systematic reviews of DF metrics

Some most relevant research work with contributions on DF metrics definitions include a set of DF metrics for event-based Shed and Shift proposed by Liu et al. These metrics cover several aspects of the load impact, including the average depth (measured in kilowatts [kW], watts per square foot [W/ft²], and percentage of baseline whole building power) and variability of demand increase or decrease during an event, ramping capability, and energy consumption impact [13]. amongst these metrics, *demand decrease intensity* (DDI) (or *shed intensity*) (W/ft²) was used to compare the field performance in three groups of buildings (medium office, large office, and retail) with prototype building simulation results. Miller and Carbonnier from the New Buildings Institute proposed a set of eight metrics for evaluating building and grid interaction [14]. These metrics are divided into two categories: (1) evaluating load shape and (2) measuring a building's potential to serve as a grid resource. The metrics are calculated from annual hourly load profile output from whole building simulations and are presented in the form of scores between 0% and 100% (higher is better) [14]. Two load shed related metrics use one-hour and four-hour as shed durations while leaving the timing of the shed events open. A subset of these metrics are being piloted with the LEED for New Construction (LEED-NC) rating systems [15].

While this paper focuses on commercial buildings DF, it is worth cross-referencing a systematic literature review study of DF metrics for residential buildings. Li et al. reviewed the recent years of literature on characterization and quantification of residential building DF and summarized metrics, methods, applications, and more [6]. Their found metrics cover six performance aspects: energy, power, cost, duration, emission, and comfort. Energy and power changes are most common in the existing metrics. Table 1 adapted from [6] shows the top five popular DF metrics found in an extensive literature review. We added a col-

umn in Table 1 to show considerations for not adopting these metrics in our proposed benchmarking framework. Li et al. pointed out that most studies were based on simulation, while less than 15% of the studies had measurements from experiments or field tests [6]. They also identified a few significant gaps in the existing literature, the first of which was establishing a common definition and performance metrics for DF of buildings that are technology and application agnostic. These important findings also apply to commercial buildings DF and inspired the contributions of this paper. In addition, the IEA's Annex 67 (ended) and Annex 82 (ongoing) are international collaboration projects focusing on "energy flexible buildings," covering both commercial and residential buildings. Annex 67 aimed to characterize energy flexibility, and one literature review by Farulla et al. related to this effort identified several existing key performance indicators (KPIs) for building flexibility [16]. Although academically orientated, these KPIs (i.e., metrics) provide some context for commercial building energy flexibility for a range of potential grid services.

1.3.2. Additional DF-related metrics

Besides the above systematic DF metrics definitions and literature review studies, there are also a number of DF related studies that measured results using various metrics specific to the purposes and needs of their own studies. In other words, those metrics used were not necessarily intended as main contributions of the study or intended as definitions for general applications. The building DF-related metrics definitions found in these relevant studies can be categorized based on their purposes and intended applications. While there are different ways to create such categories, this study tried to simplify it by using two broad categories: metrics intended to measure DF at the **system level** versus those at the **building level**. The latter is more relevant to this paper. Within each of the two categories, there may be metrics that address different aspects, i.e., subcategories. Representative, rather than comprehensive, examples are provided in Table 2; more context is also provided for both categories below. These metrics were reviewed and not considered best fit for benchmarking building DF using field measured data; the reasons are explained in Table 2. Note that studies about fast DR for grid ancillary services are not included due to our focus on load shed and load shift.

System level

- **System load impact:** Langevin et al. projected the magnitude of the technical potential of U.S. regional system peak demand and electricity generation impact from commercial and residential building-level energy efficiency and DF measures combined in 2030 and 2050 [9]. Key metrics used for impact results in this study include summer and winter daily average peak period demand reduction (gigawatts, GW) and annual electricity use savings (terawatt-hours, TWh). In addition, the above two metrics were normalized by building floor area (i.e., W/ft² and kWh/ft²) in evaluating the impact from various measure types. Zhang et al. evaluated the aggregated DR load impact from residential heat pumps using simulation with a focus on helping the grid to prepare for a "snapback" effect with increased load from restoring space temperature after a DR event (referred to as "paybacks" in the paper) [17]. The paper proposed multiple metrics for the different stakeholder groups involved, including the grid operator, electricity retailers, aggregators, and consumers. The system level power and energy related metrics were defined as ratios between the DR and the business-as-usual scenarios.
- **System cost and emission impact:** Gerke et al. studied the interactive impact of EE and DR on power system costs and emissions using whole building simulations and power system modelling to understand when residential and commercial EE and DR strategies compete with or complement each other [18].

Building level

- **Theoretical characteristics of load shape changes:** Aduda et al. proposed performance parameters for office buildings based on the

Table 1
Top Five Popular Energy Flexibility Metrics in [6].

Metrics	Definition	Performance Aspects	Need Baseline	Reasons for Not Adopted in Our Framework
Peak Power Reduction	Reduced power demand during peak hour due to flexible operation.	Power	Yes	Useful for customer bill impact, but grid events may not always align with peak hours.
Peak Power Reduction Percentage	Percentage of reduced power demand during peak hour due to flexible operation.			
Flexibility Factor	Indicator of a quantity (e.g., energy, cost, emissions) during high load hours versus during low load hours, which ranges between −1 (quantify was only during high load hours) and 1 (quantify was only during low load hours).	Energy, Cost, Emissions	No	Addresses the difference during high- and low- load periods of a building's load shape but does not measure DF performance in a grid event against a baseline. As a ratio, it also does not address how a building's flexibility contributes to the grid's needs.
Self-sufficiency	The degree to which the on-site generation is sufficient to fill the energy needs of the building.	Energy	Yes	Focuses on meeting self consumption with onsite generation; does not address load flexibility for grid service.
Self- consumption	The degree to which the on-site generation is directly consumed by the building.			
Capacity of ADR	Reduced energy consumption during an ADR event	Energy	Yes	Close to our metric although DR performance is better measured by demand reduction rather than energy reduction.
Efficiency of ADR	The energy storage efficiency during an ADR event			Specific to load shift and storage systems; considered as a secondary metric.
Flexibility Index	The fraction of saved cost from a penalty-aware operational strategy (flexible) compared with a penalty-ignorant operation strategy (baseline).	Cost	Yes	Need energy metric for DF benchmarking before layering cost on top.

Table 2
Summary of Additional Metrics in Demand Flexibility Studies.

Category	Subcategory	Metrics	Data Needed	Reasons for Not Adopted in Our Framework
System Level	System load impact	Summer/Winter daily average peak period demand savings (GW, W/ft ²), Annual electricity use savings (TWh, kWh/ft ²) [9] A set of system-level power and energy related metrics defined as ratios of DR and business-as-usual scenarios [17]	Simulated hourly building load shapes aggregated by stock Simulated hourly aggregated system load	Not building level metrics Not building level metrics
Building Level	System cost and emission impact	Peak system-level net load (MW), Max 3-hour net-load ramp (MW) [18]	Aggregated system load	Not building level metrics
	Theoretical characteristics of load shape changes	Power flexibility parameters associated with 4 theoretical phases; Operative temperature (°F), Predictive mean vote for comfort [19] Flexibility factor, absolute/relative grid support coefficient [20] A set of metrics categorized into load shifting ability, power adjustment, energy efficiency, and cost efficiency [21]	Unclear due to lack of examples; possibly simulated sub-hourly whole building power and comfort indicators	In a theoretical stage. No example of applying these metrics in a real building to show feasibility and usefulness. Counterfactual baseline accuracy can be a challenge in showing theoretical power changes.
	Simulation- based building demand and comfort impact	Demand shed intensity (W/ft ²), Indoor temperature increase (°F) [22]	Simulated hourly whole building loads and indoor temperature	(overlaps with adopted metric D1 in Section 2.1)
	Simulation- based utility bill and comfort impact	Utility cost savings (cents/ft ²), Peak demand reduction (W/ft ²), Predicted percentage of dissatisfied index (%) [23]	Simulated hourly whole building loads and comfort indicators	Need energy metric for DF benchmarking before layering cost on top. In real buildings, DF performance from grid events often happens outside peak hours.
	Field studies of DR performance	Average fan power change (kW), Average chiller power change (kW), Additional fan energy consumption (kWh), Additional chiller consumption (kWh), Additional reheat consumption (therms per hour), Effective temperature deviation (°F), Unmet cooling service (MMBtu) [24]	Hourly and sub-hourly HVAC subsystem data	Specific to HVAC related DF strategies. Requires granular measured data that is not commonly available; requires a high level of effort for calculation.

theoretical load change stages and used operative temperature and predicted mean vote to measure the impact of occupant comfort [19]. The definitions of performance parameters associated with granular stages were informed by the response time needed in various grid services. The paper did not provide an example of applying these concepts and metrics in a real building, which can be a challenge given the stochastic nature of real building power demand data and the complexity in creating a baseline demand profile. There

are also examples related to the aforementioned Annex 67 — an overview by Li and Shi covers theoretical methods for characterizing EF [20] and Johra et al. summarized existing KPIs in four categories and conducted a sensitivity analysis of the influence of several building characteristics on flexibility, focusing on load shifting ability and energy cost efficiency metrics [21]. These studies tested concepts using office buildings as examples, although they are simulation based

and fall short of exploring the complexities in real building applications.

- **Simulation-based building demand and comfort impact:** Luo et al. used surrogate models to predict the impact of several DR strategies on commercial building electricity demand and occupant comfort during a DR event window and compared model predictions with whole building simulation results [22]. The study used demand shed intensity (W/ft^2) and indoor temperature increase ($^{\circ}F$) as key metrics for demand and comfort impacts, respectively.
- **Simulation-based utility bill and comfort impact:** Cai and Braun evaluated the impact of three DR strategies (with daily optimization criteria) in seven small commercial building types in 14 U.S. locations using simulation [23]. They evaluated impact in three areas: utility cost savings (cents/ ft^2), peak demand reduction (W/ft^2), and occupant comfort impact using the predicted percentage of dissatisfied index (%). These metrics were from the utility customers' perspective under common flat and time-of-use electricity rates with or without demand charges.
- **Field studies of DR performance:** Given the relationship between DR and DF discussed in Section 1.1, DR performance measurements are directly relevant to DF metrics. Because a main contribution of this paper is benchmarking DF for real buildings, this category is particularly relevant and emphasis is given to studies with clear metrics applied to multiple buildings. Keskar et al. conducted hundreds of load-shifting DR experiments using a global temperature adjustment (GTA)³ strategy in eight university campus buildings [24]. They created metrics at the HVAC subsystem level to quantify power response and energy consumption change and reviewed impact on occupant comfort. The calculation of these metrics requires granular subsystem data from building automation systems which make them hard to be widely adopted.

1.3.3. Foundational DF metrics from field studies

One important contributor of DR field studies is the Demand Response Research centre (DRRC) at Lawrence Berkeley National Laboratory, which published a number of DR field studies between 2001 and 2015. The authors have reviewed 33 commercial building DR field studies published by the DRRC and summarized the metrics used in these publications by their frequency in Fig. 1. Several of those most worth noting deployed a set of common metrics, as shown below for quantifying the DR performance during each DR event [25, 26, 27, 28]. The energy and power related metrics are listed on the left in the figure; indoor comfort and air quality related metrics are listed on the right. Each of the field studies included one or more (up to 96) sites, with most of them being office buildings or retail stores in the United States.

The three most frequently used energy and demand related metrics found in these studies were:

- 1 Average demand reduction per event (unit: kW)
- 2 Average demand reduction relative to the total building demand (unit:%)
- 3 Average demand reduction per floor area (unit: W/ft^2 , or W/m^2)

These are also used as key metrics in DR field studies by other researchers; some examples include [29, 30, 31, 32]. Because this paper focuses on establishing metrics for benchmarking measured building DF performance, the above most common metrics found in the DR field studies became an important part of the foundation for the proposed metrics definitions presented in this paper (Section 2.1). They were considered suitable for benchmarking real buildings' DF performance because (a) they support normalized comparisons across buildings; (b)

they have been tested in several field studies or more; (c) they align with performance evaluation in existing DR programs and are therefore useful to stakeholders.

1.4. Research objectives and contribution

The earlier discussions of status quo and gaps inspired the following questions with regards to benchmarking methodology and insights to gain from available data:

- How to benchmark building DF when there are so many influential factors?
- How to create a common DF benchmarking framework that allows meaningful comparison of DF metrics measured under different conditions across buildings in different regions?
- What is the size and range of load shed from common strategies such as GTA and lighting curtailment in office and retail buildings? How much variation is there across events for the same building?
- Does load shifting from pre-cooling combined with GTA increase energy consumption?
- How does event duration and timing affect load shed from implementing GTA?

The research objectives and key contributions of this paper are twofold. First, we present a novel DF benchmarking framework based on several existing metrics in the literature as shown in Fig. 1. It is designed to address the challenges in benchmarking real buildings' performance which are subject to the influence of many factors. The proposed framework is both structured and flexible so as to enable an empirical approach to understand the influences from these factors tailored to a user's specific needs and encourage visualization for obtaining insights. The selected fundamental metrics (in Section 2.1) have simple definitions and have been tested in previous studies to be suitable for real world applications. This framework design makes it possible to compare measured DF performance of buildings across geographic areas with different climates and evolving needs of the regional/local grid (reflected in event durations, timing, weather, etc.) in a meaningful way. The proposed benchmarking framework can be applied to both real building measurements and building energy simulation results, although this paper focuses on real building DF benchmarking to explore its complexity.

Second, we showed the feasibility and scalability of the proposed benchmarking framework through applying it to a significant number of real buildings and obtaining meaningful results. We also provided various examples to show how benchmarking results can be tailored in different dimensions to provide insights about building DF. For this purpose, two case studies are presented where the proposed framework is applied to two separate commercial building groups (offices and retail) that participated in utility DR programs and implemented load shed and load shift strategies. The office building group case study contributed to understanding the energy consumption implications of load shifting in real buildings. The retail building group case study included benchmarking of more than 100 near-homogeneous buildings for load shed range, variability, and influential factors. It is a unique contribution filling in the gap of measured DF performance of larger building groups in the current literature. Based on the results, we believe applying the metrics and framework to an even larger number of buildings is going to be similar – this is an important step towards scalability. The benchmarking metrics visualization results in various dimensions of these case studies provided insights into the above questions; moreover, they served as examples of how others can use this framework to benchmark buildings' DF performance in other building-level evaluation and system-level planning applications (Section 5.1). The targeted stakeholders for these applications are building managers/operators/practitioners, aggregators, utilities, and grid operators.

The remaining sections of this paper are organized as illustrated in Fig. 2.

³ GTA is raising zone temperature setpoints “globally” across a number of zones in the building during a Shed event. It is common for buildings to couple it with “pre-cooling” by lowering the zone temperature setpoint for some time before the Shed in order to achieve deeper demand shed and provide better comfort during the event.

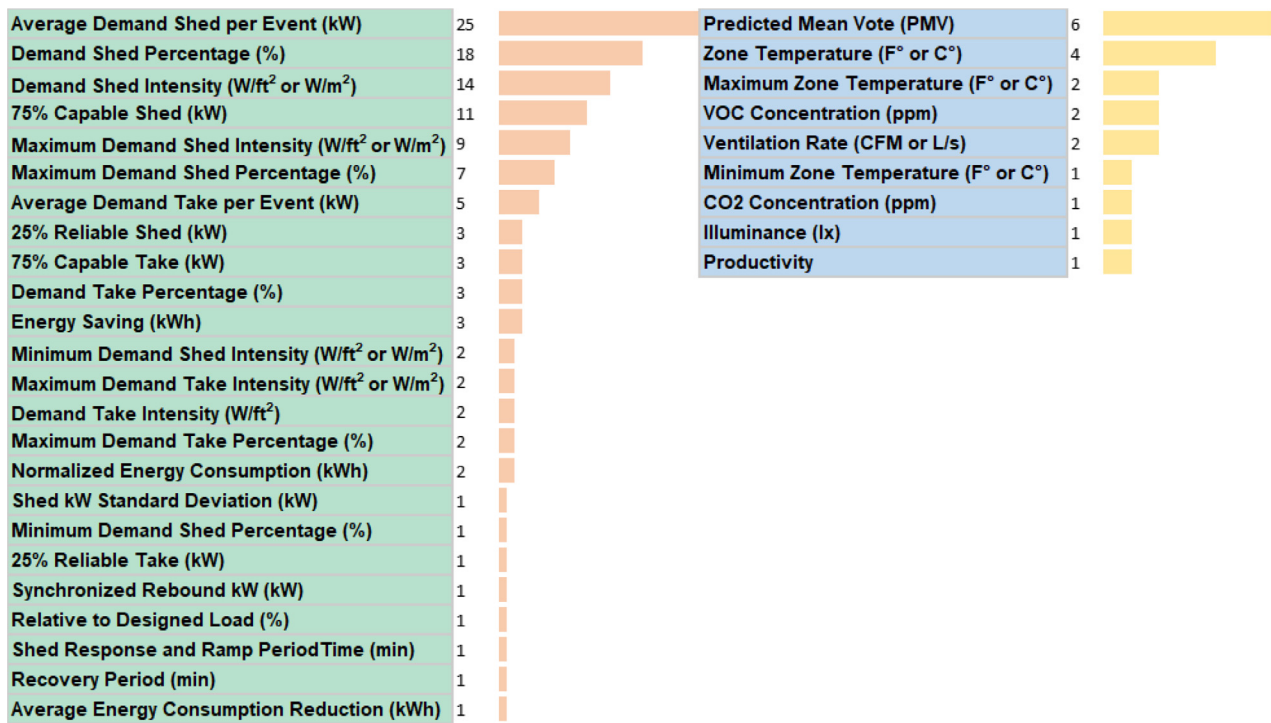


Fig. 1. Metrics Used in DR Field Studies, Ranked by Frequency.

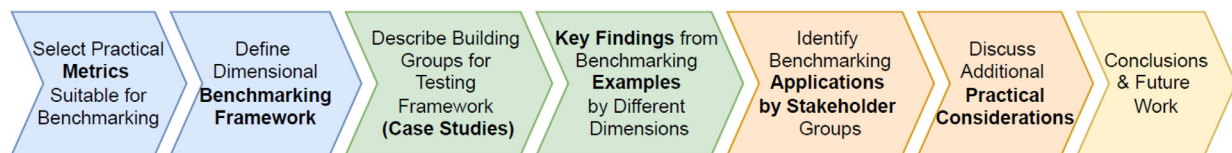


Fig. 2. Flow Chart of Contents Structure in This Paper.

2. Methodology (Metrics definitions and benchmarking framework for shed and shift)

In this paper, we define a set of metrics for “load shed” (Shed) and “load shift” (Shift). It is worth noting that load shedding and load shifting results can be achieved through two broad types of approaches: (1) DR event signals and (2) influencing load shape through regular or continuous signals such as prices or carbon emissions indicators. The metrics defined in this paper focus on the former — event-based dispatches — and apply to traditional building loads. For buildings with distributed energy resources (DERs) such as energy storage, photovoltaics, or other self-generation sources, these metrics and their calculation methods will need to be adapted, and that is discussed in the Conclusions and Future Work section.

Considering this framework focuses on field DR performance, it is assumed that the implemented strategies would meet a building’s services and occupant’s comfort requirements. For example, the GTA strategies used in the case studies adjust space temperature setpoints by only a few degrees and stay within the ASHRAE Standard 55 recommended thermal comfort zones. Therefore, comfort is not part of the defined metrics.⁴ This framework is not intended for evaluating trade-off between potential pay-off and building service compromises.

Fig. 3 illustrates the structure of the DF benchmarking metrics framework for Shed and Shift. We first selected a set of existing metrics to characterize a building’s DF performance during a single event; they are well suited for comparison across buildings on a normalized basis.

These “single-event” metrics are a foundational building block for the benchmarking metrics. Each single-event metric has a set of Attributes, which are used to organize and obtain the focus of a given benchmarking exercise. In other words, the benchmarking metrics are aggregated from single-event metrics through the lens of a given set of Attributes. In addition, in benchmarking a given building, the selection of a similar cohort may consider several factors, as shown in the figure. One difference between the “cohort factors” and the “single-event metric attributes” is that the former are generally static characteristics of a building such as geographic location, construction, or operation related; the latter are associated with and can vary by each event. Both of these factors and attributes are not exclusive lists and may be used to filter and organize the benchmarking cohort and analysis dimensions such that the results are tailored to specific needs. More detailed definitions of single-event metrics, metrics attributes, and benchmarking metrics are provided in the following sections (2.1, 2.2, and 2.3). Section 2.4 illustrates the steps involved in the benchmarking metrics calculation procedure. Section 2.5 discusses the baseline method used in the case studies in this paper.

2.1. Single-event metrics definitions

In this section, we standardize the definitions of several existing key metrics (as shown in Table 3 and Table 4) to quantify DF associated with a single Shed or Shift event. The definitions of these single-event metrics draw from established practices in existing DR programs and frequently used field-measured DR performance metrics in literature, as summarized in Fig. 1. These metrics allow building managers, operators, and others to understand a building’s ability to shed or shift loads on

⁴ Although important, it is often impractical to obtain occupant comfort related data from real buildings.

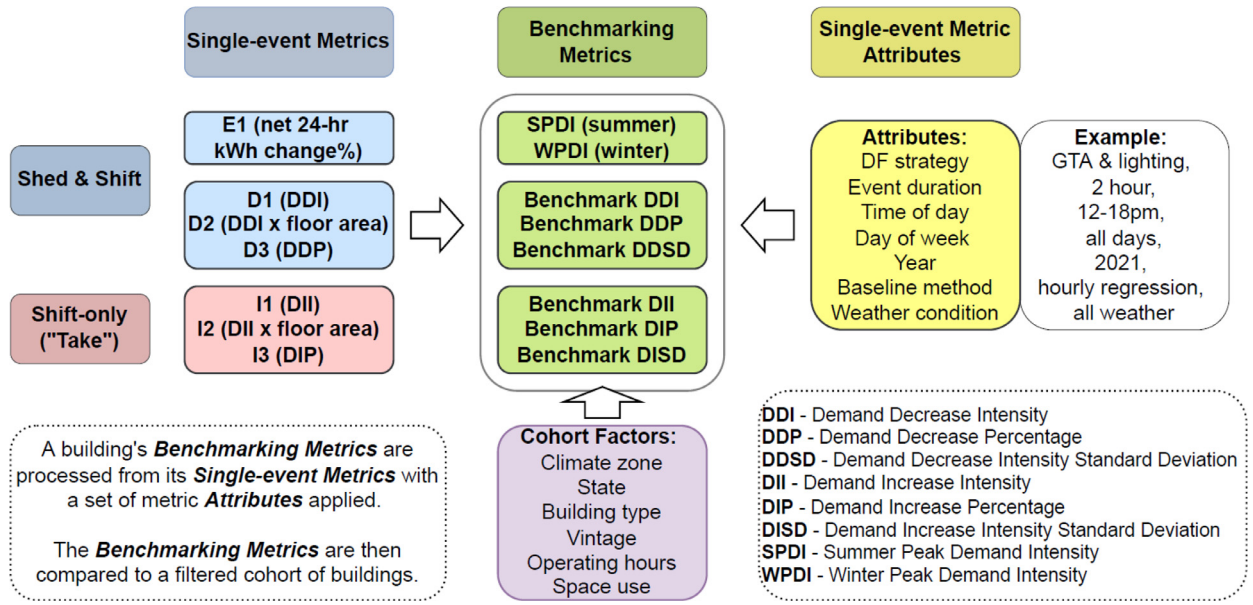


Fig. 3. DF Benchmarking Metrics Framework with Three Building Blocks: Single-event Metrics, Metrics Attributes, and Benchmarking Metrics.

Table 3
Single-event Metrics for Demand Decrease (Load Shed).

Metrics	Formula / Definition	Unit
E1: Net Building Consumption Change Percentage (24 h)	= Net daily kWh change / Baseline daily kWh consumption x 100%	%
D1: Demand Decrease Intensity (DDI)	= D2 / Floor Area	W/ft ² (or W/m ²)
D2: Demand Decrease	= Average demand decrease during a single "shed" period	kW
D3: Demand Decrease Percentage (DDP)	= D2 / Baseline average demand during "shed" period	%

Table 4
Single-event Metrics for Demand Increase (Load Take).

Metrics	Formula / Definition	Unit
I1: Demand Increase Intensity (DII)	= I2 / Floor area	W/ft ² (or W/m ²)
I2: Demand Increase	Average demand increase during a single "take" period	kW
I3: Demand Increase Percentage (DIP)	= I2 / Baseline average demand during "take" period	%

a first order basis; that is, to focus on both the absolute and relative magnitude of the load change over a specific period of time. Calculating these metrics requires hourly or sub-hourly interval metered data or simulation outputs for building total power demand (kW).

Three metrics are defined to measure the average demand decrease during a Shed event: *demand decrease intensity* (W/ft², or W/m²) (D1), *demand decrease* (kW) (D2), and *demand decrease percentage* (%) (D3), as shown in Table 3. D2 is an intermediate step in calculating D1 and D3, and is useful in measuring the load shed results for performance based compensation. D1 is a useful metric because normalizing by building floor area allows comparison across buildings and benchmarking. D3 describes the load shed relative to the baseline total building load during the same period, which can also be used as a benchmarking metric. The calculation of these three metrics all require a baseline that represents a building's counterfactual load profile on an event day if no DF strategy were deployed. It is worth noting that, as Fig. 1 reveals, a floor area normalized metric such as "demand shed intensity" (same as D1) is particularly useful in comparisons across building types and climates.

Load shift consists of a two-part load change: demand increase ("load take," or Take) and demand decrease (Shed). Therefore, all of the above metrics defined for load shed also apply to the Shed part of load shift. In addition, an additional set of three metrics — I1, I2, and I3 — were defined for "load take" (in Table 4). These are similar to D1, D2, and D3 for Shed. They are also calculated using the same baseline load profile. In the metrics definitions above, demand decrease is defined as positive values, so as to be consistent with the conventions in traditional DR programs, and therefore, demand increases are negative values.

In order to understand how a load shed or shift strategy affects energy consumption, another metric, E1, is defined; it measures the building's net energy consumption change from baseline energy consumption as a percentage in a 24-hour window around the shed or shift event. It assumes that typical load shed or shift events are completed within 24 hours but can be modified for different needs. *Net energy consumption increase* is defined as positive values.

It is worth noting that no separate metrics are defined for ramping capability. The authors believe that ramping service is best provided by buildings in aggregate rather than individually. Counterfactual baselines often introduce significant errors; therefore, it is not advisable to measure ramping at the individual building level. Instead, aggregators can stagger the start time of load shed or take from a number of buildings to achieve ramping results. And, the above metrics measuring average load shed or take for individual buildings can support such planning.

2.2. Single-event metric attributes and cohort factors

In the presented DF benchmarking framework, collecting single-event metrics data is a foundational step. Taking EUI as a benchmarking metric example again, a weather normalized EUI is sometimes used to exclude the impact from year-to-year weather variations [33]. However, building DF metrics are more complex because electricity use is dynamic, varying by day of week, hour of day, season, and weather. It is common to observe significant variations in the size of load shed across DR events for the same building. Therefore, it would not be meaningful to use a single-event metric value to represent a building's DF ability without specifying a number of key conditional parameters around it. To address this issue, we propose a set of *Single-event Metric Attributes* (Attributes) (as shown in Fig. 3) to cover these key conditional parameters; they are not intended to be comprehensive but are considered essential. For example, a DDI (D1) metric value of 0.5 W/ft² (5 W/m²)

may be associated with the following Attributes values to illustrate the context. This concept applies to all of the single-event metrics shown in Table 3 and Table 4.

- **DF strategy:** turning off 50% of the lights and implementing a 3°F (1.7 °C) GTA
- **Event duration:** 2 hours
- **Time of day:** 12–6 pm
- **Day of week:** every day
- **Year:** measured in 2021
- **Baseline method:** hourly weather regression
- **Weather condition:** peak outside air temperature (OAT) = 82°F (28 °C)

In addition to the above Attributes, there are another set of factors associated with a **building's characteristics** which have significant influence over its DF, such as geographic location, space use, construction, and operation. Unlike the Attributes, these factors are generally static, as opposed to varying event by event, and they are useful in identifying the right cohort of buildings for benchmarking. Therefore, we call them *Cohort Factors*, as shown in Fig. 3. Although they tend to be static, they are subject to changes when major retrofits or renovations happen.

In addition to the Attributes and Cohort Factors, there is a third category – **stochastic factors** – that influence a building's DF. A building's occupancy and internal loads are stochastic by nature, as are changes in business operations, and all can significantly influence loads and DF for many types of buildings. Commissioning issues of energy-consuming equipment and systems can have profound impacts on DF, yet they are often overlooked.

In summary, a wide range of factors can significantly influence a building's DF, and they can be organized into the following three categories:

- 1) **Building characteristics (cohort factors)**, such as building type, vintage, operating hours, space use, climate zone, and state⁵
- 2) **Single-event metric attributes**, such as DF strategy,⁶ event duration,⁷ time-of-day,⁸ day-of-week, year,⁹ baseline method,¹⁰ and weather condition¹¹ (e.g., daily peak OAT)
- 3) **Stochastic factors**, such as stochastic phenomenon in occupancy, internal loads, and business operations, as well as commissioning issues in equipment and controls

amongst those three categories, the Single-event Metric Attributes are a focus of this paper. Each Attribute can be used as a *Dimension* in vi-

⁵ For some datasets and applications, independent system operators or utilities may be suitable alternatives to the state.

⁶ A building may support multiple control strategies to shed or shift load, such as adjusting thermostat setpoint to reduce HVAC load, dimming lights, curtailing plug loads, or discharging thermal or electrical storage. Not all of the available strategies may be used in all Shed/Shift events, as they can be prioritized based on the building's utility tariff, utility program rules, and the impact on building services. For example, a building may choose to deploy a single strategy for an economic program versus several strategies in an emergency DR program event.

⁷ Electrical load shed from some building loads are easier to sustain than the others. For example, it is easier to dim lights for a few hours than to cycle off HVAC for hours because the space may get uncomfortable for the occupants when the HVAC is off.

⁸ Building loads (e.g., lighting, plug load, HVAC) and their ability to shed or shift vary throughout the day and week as occupancy and operation mode change (e.g., HVAC setback during unoccupied periods). Time-of-day can be a category rather than a value, e.g., 0:00–8:00, 8:00–12:00, 12:00–18:00, 18:00–0:00.

⁹ The same building's DF performance from the same DF strategy can change significantly over time due to operational changes, equipment conditions, and other factors.

¹⁰ DF metric value can vary significantly depending on the chosen baseline method. See more discussion in Section "2.5 Baseline Method."

¹¹ Some building loads, such as HVAC, are often dependent on weather conditions, and therefore can influence load shed or shift from these loads.

sualizing a given building's DF performance. Then this Attribute can be referred to as an *Dimension-Attribute*. In other words, we are interested in identifying trends in the measured DF against a chosen Dimension-Attribute and any potential correlations between them. Section 2.4 illustrates how benchmarking metrics may be visualized by different Dimensions.

2.3. Benchmarking metrics definitions

In this section, the authors propose definitions of the eight DF benchmarking metrics for load shed and shift in Table 5. All of these metrics are presented in the form of either intensities (normalized by floor area) or percentages of total building load, to allow comparison across buildings.

As discussed in Section 2.2, considering that many variables influence a building's DF metrics measured in each *Shed* or *Shift* events, metric values used for benchmarking a building's DF against other buildings should be presented with a set of conditional parameters, i.e., the Attributes defined in Section 2.2. Once a set of Attributes are determined for benchmarking a group of buildings, the benchmarking metrics of each building can be calculated according to Table 5, based on its single-event metrics.

The eight benchmarking metrics shown in Table 5 are presented in three categories: (1) peak demand, (2) load shed, and (3) load take. Below are some considerations behind these definitions. First, a building's ability to shed or shift load should be evaluated in the context of the building's peak demand in order to show its relative significance. A building's peak demand is often correlated to the outdoor temperature — i.e., electricity demand tends to peak on the hottest and/or coldest days of the year depending on whether the building uses electricity for both space cooling and heating. Therefore, two normalized metrics — “summer peak demand intensity (W/ft^2 , or W/m^2)” (SPDI) and “winter peak demand intensity (W/ft^2 , or W/m^2)” (WPDI) — are proposed to provide reference for cooling and heating season peak demands.

The remaining six metrics for Shed and Take are all defined around the five events that yielded the highest load shed in the cooling or heating season in a year, i.e., the “top-5 shed.” This design considers that although absorbing renewable generation by load take is sometimes needed, load shifting is often intended to alleviate either system peak conditions or steep ramping, both of which require load shed. In other words, Shed is more critical than Take in our proposed framework. One consideration for using the average of the top-5 sheds is to reduce influence from random factors in a building's DF performance. Data availability is another consideration; current DR programs in the U.S. often limit the number of events in a year to somewhere between 12 and 16 events, and it is common for buildings to opt out of some events. Therefore, performance data from five participated DR events¹² is intended to balance data availability with representativeness of a building's average performance.

Benchmark Demand Decrease Intensity (DDI) and *Benchmark Demand Decrease Percentage (DDP)* measure the size of load shed. *Benchmark Demand Decrease Intensity Standard Deviation (DDSD)* reflects the consistency in the load shed performance across those five events. The benchmarking metrics for Take are similar to Shed. These benchmarking metrics values can be established for cooling versus heating seasons. This paper presents two case studies that applied these benchmarking metrics to evaluate the performance of real buildings when participating in DR events in the cooling season.

¹² When there are fewer than five event data points that meet a certain set of Attributes criteria, these benchmarking metrics can still be calculated using fewer data points, although the results will be less robust; we do not recommend calculating these metrics using fewer than three events.

Table 5
Benchmarking Metrics for Shed and Shift.

	Benchmarking Metrics	Formula / Definition	Unit
Peak (for reference)	SPDI: Summer Peak Demand Intensity	= Summer hourly (or 15-min) peak demand / Floor area	W/ft ² (W/m ²)
	WPDI: Winter Peak Demand Intensity	= Winter hourly (or 15-min) peak demand / Floor area	W/ft ² (W/m ²)
Shed	Benchmark DDI ¹ (Benchmark Demand Decrease Intensity)	= Average of the five highest per-event D1 ² values ("top-5 shed")	W/ft ² (W/m ²)
	Benchmark DDP ¹ (Benchmark Demand Decrease Percentage)	= Average of the per-event D3 ² values of the "top-5 shed"	%
	Benchmark DDS ¹ (Benchmark Demand Decrease Intensity Standard Deviation)	= Standard Deviation of the "top-5 shed"	W/ft ² (W/m ²)
	Benchmark DII ¹ (Benchmark Demand Increase Intensity)	= Average of the I1 ³ values associated with the "top-5 shed"	W/ft ² (W/m ²)
Take (apply to Shift only)	Benchmark DIP ¹ (Benchmark Demand Increase Percentage)	= Average of the I3 ³ values associated with the "top-5 shed"	%
	Benchmark DISD ¹ (Benchmark Demand Increase Intensity Standard Deviation)	= Standard Deviation of the I1 ³ values associated with "top-5 shed"	W/ft ² (W/m ²)

¹ These benchmarking metrics values are associated with a given set of *Attributes* values such as the example shown in Section 2.2. A separate set of metrics may also be used for the cooling and heating seasons.

² See Table 3.

³ See Table 4.

2.4. Calculating and visualizing benchmarking metrics

2.4.1. Visualizing benchmarking metrics by different dimensions

Section 2.1 through 2.3 have covered all of the building blocks for benchmarking building DF, and these are illustrated in Fig. 3. This section discusses how to structure a benchmarking task based on the previous building blocks and illustrates how to visualize benchmarking metrics for a given purpose.

A key concept explained in Section 2.2 was that each Attribute can be a Dimension in analysing trends in DF benchmarking metrics. Depending on the purpose of the benchmarking task and nature of the dataset, zero, one, or two Dimension-Attributes may be selected for the following types of benchmarking:

- **Zero-dimensional (0-D):** The buildings in the benchmark group share the same values for all seven *Attributes*.
- **One-dimensional (1-D):** One of the *Attributes* has multiple values, and it is of interest to understand its influence by minimizing influence from other *Attributes*. If understanding the influence of multiple *Attributes* separately is of interest, then conducting one-dimensional benchmarking one-by-one could achieve this.
- **Two-dimensional (2-D):** Two of the *Attributes* have multiple values, and it is of interest to understand their combined influence.

Fig. 4 shows an example for each of the above benchmarking types. In each example, the selected Dimension-Attribute(s) is in red text; the other *Attributes* are used as filters to be applied on the single-event metrics. It is important to record the *Attributes* values for each single-event metric, to allow comparisons by different dimensions using different filters as needed in various benchmarking applications.

In general, zero-dimensional and one-dimensional benchmarking are expected to be the most commonly used. Section 4 presents one zero-dimensional example and a few one-dimensional examples. Given the complexity in measured real building performance, it can be difficult to achieve meaningful results with two-dimensional benchmarking. Nonetheless, an example is provided to show how visualization may be used for such more sophisticated benchmarking.

2.4.2. Benchmarking metrics calculation procedure

The procedure of calculating a building's DF benchmarking metrics for visualization generally involves the steps illustrated in Fig. 5. This illustration is most typical for one-dimensional benchmarking. A set of DF metrics *Attributes* should be first identified based on the need of a given benchmarking task. Such examples are provided in Fig. 4. This

selected set of *Attributes* values will be used to filter out any nonconforming events. The values of a Dimension-Attribute will be applied one at a time in the filtering process with a loop structure. Within the loop, single-event metrics (e.g., D1, D3) will be calculated for all conforming events after filtering. Then the five events with the highest D1 values will be selected for calculating the benchmarking metrics (e.g., Benchmark DDI, DDP, and DDS). And such a process will be repeated for each value of the Attribute serving as the Dimension. In the example shown in Fig. 5, the Benchmark DDI will be calculated for each shed duration value (1, 2, ..., 6 hours) for each building in the benchmarking group. At the end, the benchmarking metrics results will be visualized in different ways to provide different insights, as illustrated earlier, in Fig. 4.

2.5. Calculating baseline load profile

A baseline load profile is required for calculating the single-event metrics; it represents a building's counterfactual load profile on an event day if no DF strategy were deployed. Although selecting a baseline method is independent from the metrics definition, it can have significant influence on the calculated metric values and is therefore included as a single-event metric Attribute. While it is outside the scope of this paper to compare existing baseline methods, this section describes the baseline method used in the two case studies described in Section 3.

These case studies employed a baseline method that is based on an hourly regression model using a single variable (outdoor air temperature, or "OAT"); it has been tested in multiple previous DR field studies [34, 35]. Specifically, it involves constructing a linear change-point regression model for each hour in a day (weekdays and weekends are treated the same). The hourly models are developed using 15-minute whole building power data of the 14 days prior to each DR event excluding event days and holidays. The chosen baseline method was evaluated for its accuracy along with the conventional "10-in-10"¹³ and "10-in-10 with adjustment"¹⁴ methods as well as two variations of the chosen method itself (longer model training period and adding day of the week as another variable) using two error metrics from ASHRAE Guideline

¹³ "10-in-10" baseline: a type of day-matching baseline method which takes the average demand during DR event hours over the previous 10 eligible baseline days (excluding weekends, holidays, DR event days, and none-operation days)

¹⁴ An adjustment factor is a ratio constructed by comparing the average demand during the first 3 of the 5 hours before a DR event start time on a DR event day vs. the 10 eligible baseline days. The adjustment is limited to $\pm 20\%$.

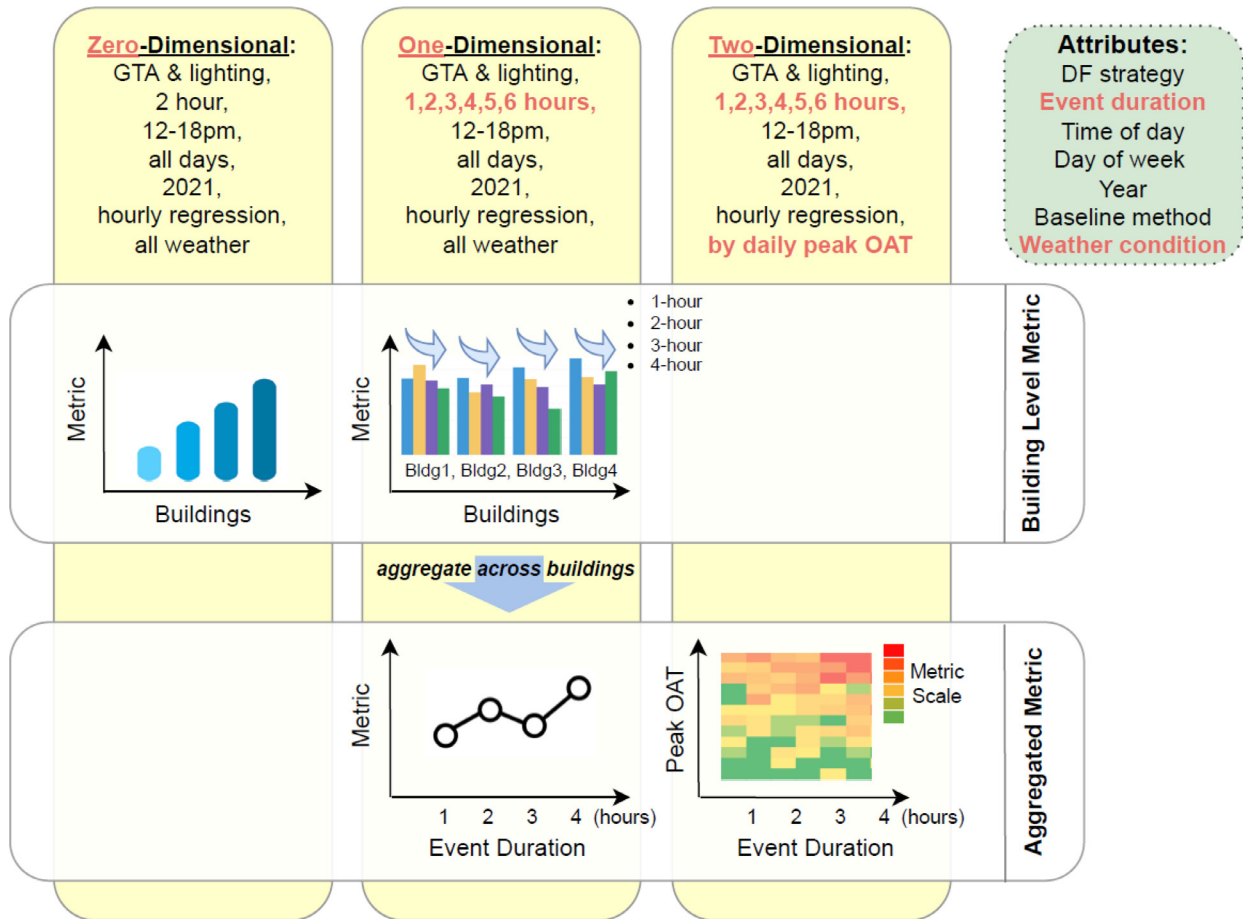


Fig. 4. An Example of DF Benchmarking Metrics in Different Dimensions.

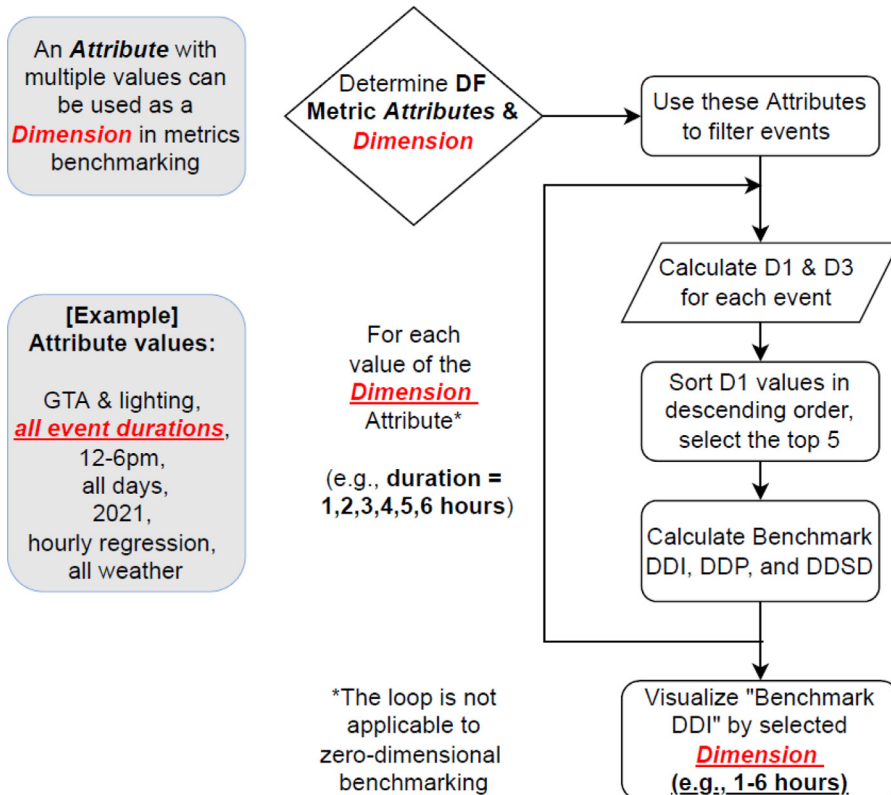


Fig. 5. DF Benchmarking Metrics Calculation Procedure Diagram.

Table 6
Characteristics of 11 Medium Offices in California That Participated in DR Events.

Site ID	Building Use	Floor Area (ft ²)	Floor Area (m ²)	Year Built	Construction	HVAC System	DR Strategy
A	Office	68,955	6406	1990	Concrete	RTU + VAV boxes	Pre-cooling+GTA
B	Office	62,800	5834	1988	Concrete	RTU + VAV boxes	GTA
C	Office	38,808	3605	1988	Concrete	RTU + VAV boxes	Pre-cooling+GTA
D	Office	73,730	6849	1988	Concrete	RTU + VAV boxes	GTA
E	Office	70,069	6509	1993	Steel	RTU + VAV boxes	GTA
F	Office+Classroom	80,750	7501	2001	Steel	RTU + VAV boxes	Pre-cooling+GTA
G	Office+Data centre	104,501	9708	2005	Steel	RTU + VAV boxes	Pre-cooling+GTA
H	Office+Classroom	119,035	11,058	2002	Steel	RTU + VAV boxes	Pre-cooling+GTA
I	Office+Auditorium	81,079	7532	1994	Concrete	RTU + VAV boxes	GTA
J	Office	112,717	10,471	1990	Steel	Central Chiller + VAV boxes	GTA
K	Office	83,698	7775	2004	Concrete	RTU + VAV boxes	Pre-cooling+GTA

14 [36] – Normalized Mean Bias Error (or “nMBE”) and Coefficient of Variance of the Root Mean Square Error, or “CV(RMSE)”. The details of the method and results of the above evaluation will be published separately in [37]. It is worth noting that the chosen hourly regression baseline method intends to address the limitation of the conventional “10-in-10 with adjustment” method (commonly used in today’s utility DR programs) in load shifting applications.

3. Benchmarking case studies

This paper experiments with applying the above proposed DF benchmarking framework to two separate groups of real buildings that participated in DR programs. This section describes these building groups and shows examples of calculating single-event metrics for load shed and load shift. The single-event metrics for all DR events are then aggregated into benchmarking metrics following the procedures laid out in Section 2.4.2. Several examples of the visualized benchmarking metrics are found in Section 4, “Results.”

3.1. Medium office buildings

The first group of buildings are 11 medium office buildings located in Southern California. The building power data are from a previous DR field study [38], although a different baseline method is applied for the purpose of this paper (see Section 2.5). The basic information for each building and the deployed DR strategy are found in Table 6. The most common HVAC systems amongst these buildings are packaged rooftop units (RTUs) with variable air volume (VAV) terminal boxes. These buildings all participated in the same utility’s DR events in 2008. The DR event window was 12–6 pm (6-hours), during which time GTA strategies (2–3°F [1.1–1.7°C]) were employed for all the zones. Because five of the 11 buildings implemented a pre-cooling strategy in the morning, Shift metrics can be calculated for these buildings.

Today’s DR communication architecture is typically one-way communication, i.e. the building receives dispatch or electricity price based signals from a grid-side entity: directly from the utilities, or through third-party aggregators, or through curtailment service providers (CSP). As shown in Fig. 6, there are three common DR communication and control pathways between the electrical grid and building loads equipment. The DR implementation in this case study follows Path (a), where the utility OpenADR¹⁵ Server sends grid event signals to an OpenADR Client that resides in a building DR gateway. Upon the receipt of a grid event signal via an OpenADR¹⁶-compliant communication gateway (e.g., translates the signal and forwards it to a DR load controller),

¹⁵ OpenADR is an open, highly secure, and two-way information exchange model and Smart Grid standard. <https://www.openadr.org/>

¹⁶ OpenADR is an open, highly secure, and two-way information exchange model and Smart Grid standard. <https://www.openadr.org/>

Table 7
Example Single-event Metrics of an Office Building Deployed Load Shift on a DR Event.

Metrics	Formula/Definition	Unit	Value
Shed			
E1: Net Building Consumption Change Percentage (24 h)	Net daily kWh change / Baseline daily kWh consumption	%	-0.1%
D1: Demand Decrease Intensity	D2 / Floor area	W/ft ² (W/m ²)	0.30 (3.2)
D2: Demand Decrease Percentage	Average demand decrease during the “shed” period	kW	31.6
D3: Demand Decrease Percentage	D2 / Baseline average demand during the “shed” period	%	7.4%
Take			
I1: Demand Increase Intensity	I2 / Floor area	W/ft ² (W/m ²)	-0.26 (-2.80)
I2: Demand Increase Percentage	Average demand increase during the “take” period	kW	-27.3
I3: Demand Increase Percentage	I2 / Baseline average demand during the “take” period	%	-8.5%

the DR controller will then trigger a set of pre-programmed control sequence (e.g., GTA) in the building management system (BMS) to shed the building HVAC load. In this case study, different GTA control sequences (linear, step and exponential methods for thermostat setpoint adjustment) were simulated for each building using a validated EnergyPlus model [27] to reduce building loads during the DR event time-window. The complex relationship between GTA setpoint (degrees) and building load response was simplified using a reduced-order model, i.e., the GTA setpoint used in a given event is decided as a function of time of day and outdoor temperature, which was predetermined based on mapping of GTA setpoint and building demand decrease level from a calibrated simulation model using field measurements [28].

Fig. 7 shows an example of the 15-minute load profile of one of these office buildings that deployed pre-cooling and GTA on a DR event day. It also shows a counterfactual baseline load profile constructed per Section 2.5 and the demand change profile¹⁷ (i.e., actual demand subtracted from the baseline demand). Table 7 shows all of the single-event metrics calculated for this example event day.

3.2. Big-box retail buildings

The second group of buildings are 121 large big-box retail stores located in 10 states spread across the United States. There are more sites in some states such as Ohio, New York, Texas, and California. These stores belong to a retail chain and share similar space uses, which makes them a good cohort for DF benchmarking. The stores are generally 100,000 square feet or larger in size and were constructed before 2010 with con-

¹⁷ Shed is shown as positive values and Take as negative values, per Section 2.1.

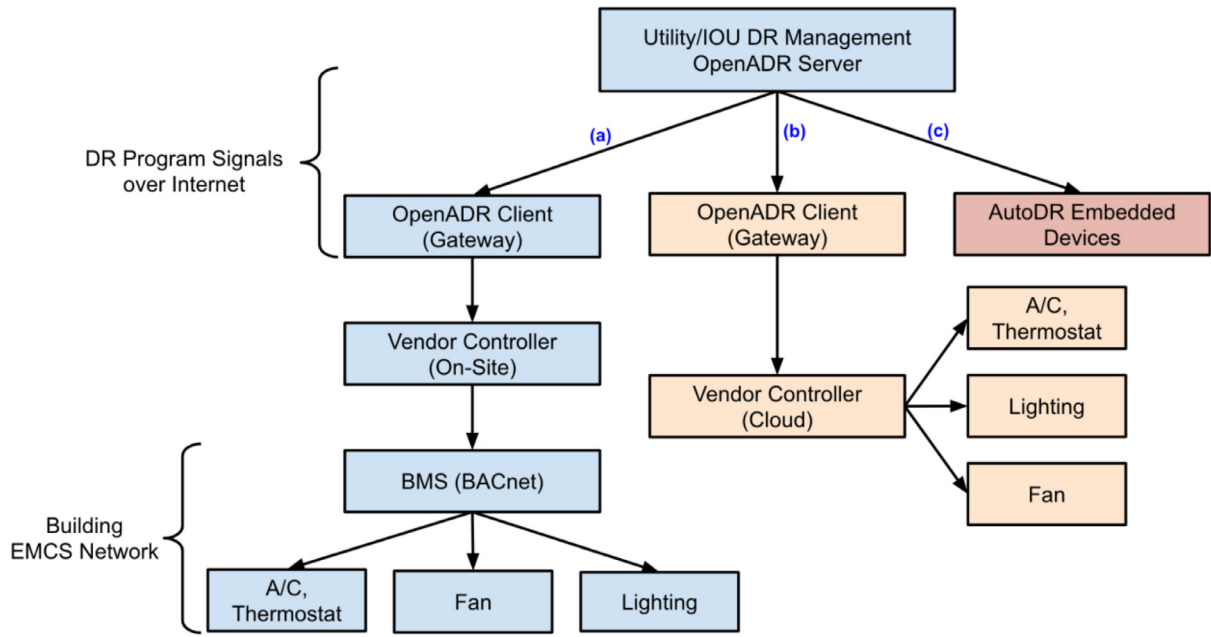


Fig. 6. Three DR Communication and Control Pathways between the Grid and Building Loads.

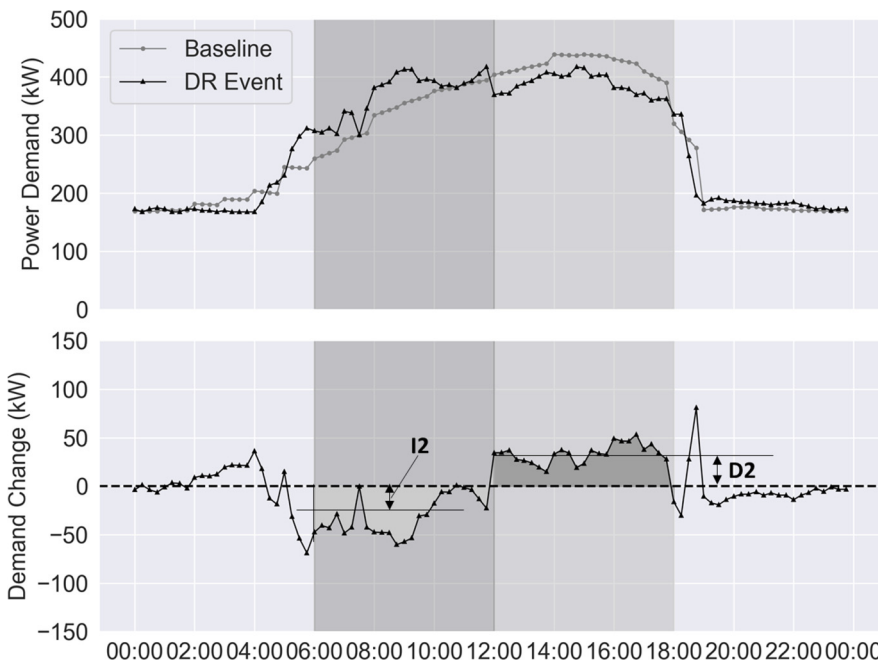


Fig. 7. Example Baseline vs. Actual Load Shapes of an Office Building That Deployed Load Shift Strategy on a DR Event Day.

crete walls and small window areas. The most common HVAC systems used at these stores are packaged rooftop units with variable speed fans.

The DR implementation in these retail stores also followed Path (a) in Fig. 6 as was in the above office buildings example although there were more nuances in the implementation of DR controller and BMS in order to accommodate varying technical capabilities of utilities, program administrators and curtailment service providers and their adopted DR event dispatch methods. Typically, upon the receipt of a grid event signal via an OpenADR-compliant communication gateway, the DR controller will trigger a set of pre-programmed control sequences to shed HVAC and lighting loads. This include: (1) about half the lights were turned off in the sales areas (the actual percentage of lights may vary by site), and (2) a number of the non-critical RTUs deployed a 3°F GTA

strategy without pre-cooling the spaces. Therefore, the Shed metrics were applied. The Shed metrics were calculated using 15-minute interval data, DR event windows, and basic building information such as floor area and location. The DR event duration and start time varied by site and by event; they were generally between one and six hours long. This presented an opportunity to benchmark this group of buildings by zero, one, and two dimensions; examples are shown in Section 4.

Fig. 8 shows an example of the 15-minute load profile of one of the retail stores that deployed lighting curtailment and GTA on a two-hour DR event day. It also shows a counterfactual baseline load profile constructed per Section 2.5, as well as the demand change profile. Table 8 shows all of the single-event metrics calculated for this example event day.

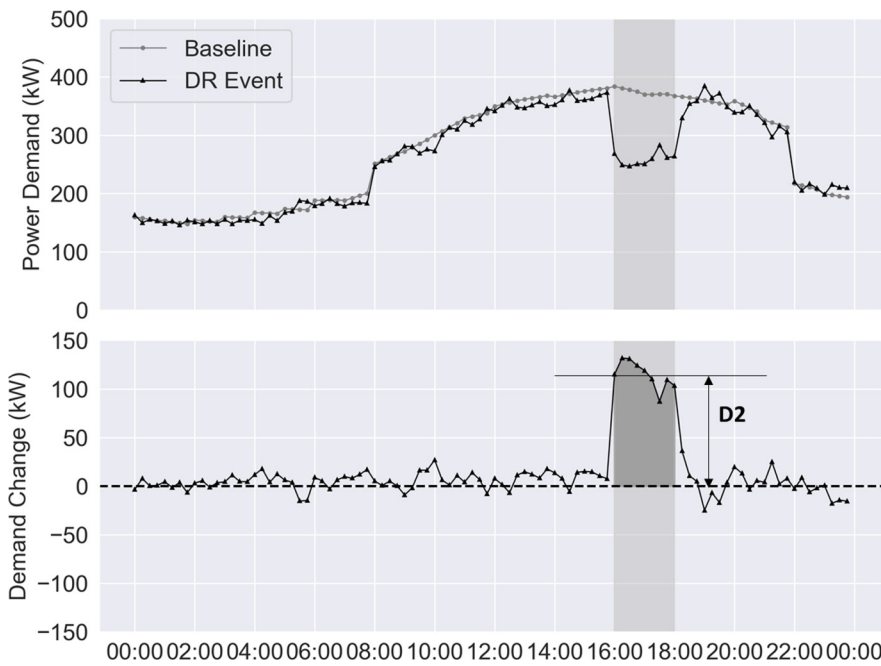


Fig. 8. Example Baseline vs. Actual Load Shapes of a Retail Building That Deployed Load Shed Strategies on a DR Event Day.

Table 8

Example Single-event Metrics of a Retail Building Deployed Load Shed on a DR Event.

Metrics	Formula/Definition	Unit	Value
E1: Net Building Consumption Change Percentage (24 h)	Net daily kWh change / Baseline daily kWh consumption	%	-3.1%
D1: Demand Decrease Intensity	D2 / Floor area	W/ft ² (W/m ²)	0.85 (9.2)
D2: Demand Decrease	Average demand decrease during the “shed” period	kW	114.7
D3: Demand Decrease Percentage	D2 / Baseline average demand during the “shed” period	%	30.7%

4. Results

This section shows various visualizations of benchmarking metrics for the two groups of buildings introduced in Section 3. As mentioned earlier, these two groups of buildings present Shift and Shed benchmarking examples, respectively. The office buildings were served by the same utility and participated in the same DR events; therefore, zero-dimensional benchmarking is appropriate. The retail stores experienced much more varying DR events, and therefore this paper explores zero, one, and two dimensional benchmarking for this group.

An important objective for this section is to show that applying the proposed benchmarking framework to real buildings for meaningful results is feasible, and to provide various example ways benchmarking results can be visualized in different dimensions to provide different insights. The zero-dimensional benchmarking results are mainly used to show the magnitude and variation of DF metrics in real buildings. In the one- and two-dimensional benchmarking examples, we also interpreted the trends in the results to the extent that our knowledge of these buildings could support. It is not in the scope of this paper to fully interpret nuances in the results at a building-by-building level, due to the limited information available.

4.1. Zero-Dimensional (0-D) benchmarking

As introduced in Section 2.4, when a group of buildings share a set of identical (or similar) *Attribute* values, it is called a *zero-dimensional*

DF benchmarking. This is the most simple form of DF benchmarking, where the *Attribute* values are identical so that the metrics values for different buildings are directly comparable.

4.1.1. Office buildings example

In the medium office buildings benchmarking example, the 11 buildings share the following *Attribute* values:

- **DF strategy:** Implementing a 2°F–3°F (1.1°C–1.7°C) GTA, and pre-cooling (for some buildings).
- **Event duration:** 6 hours
- **Time of day:** 12–6 pm
- **Day of week:** every day
- **Year:** measured in 2008
- **Baseline method:** Hourly weather regression
- **Weather condition:** hot summer days

Section 3.1 showed an example of calculating the single-event metrics for one of the office buildings. Based on these, the benchmarking metrics for each building can be derived, per Table 5. Fig. 9 shows the following benchmarking metrics for these 11 buildings as applicable: Benchmark DDI, Benchmark DDP, Benchmark DDS, Benchmark DII, Benchmark DIP, Benchmark DISD, and SPDI. The WPDI was not calculated because electricity demand data were only available for the summer season for this building group. In Fig. 9, the buildings are divided into two groups: those that deployed pre-cooling in addition to GTA and those that did not; the buildings within each group are ranked in ascending order by their Benchmark DDI. Fig. 10 shows the distribution of all Single-event D1 values in 2008 for each building. The mean value (shown as a dot) of each box matches the Benchmark DDI in Fig. 9.

In terms of load shed magnitude, the Benchmark DDI metric values for either group are similar, in the range of 0.2–0.9 W/ft² (2–10 W/m²); the Benchmark DDP metric values are in the range of 5%–25%. The range of single-event D1 values (i.e., DDI) is significantly narrower for the group of buildings with pre-cooling compared to the other group, which employed only GTA; it is about 0–0.8 W/ft² (0–9 W/m²) versus 0–1.3 W/ft² (0–14 W/m²). This reveals that the pre-cooling strategy may potentially improve the consistency of load shed although this is limited to the number of buildings in this example. In addition, a couple of buildings (#8 and #11) have a significantly wider spread of D1 values amongst the five highest D1 values than the other buildings. This effect

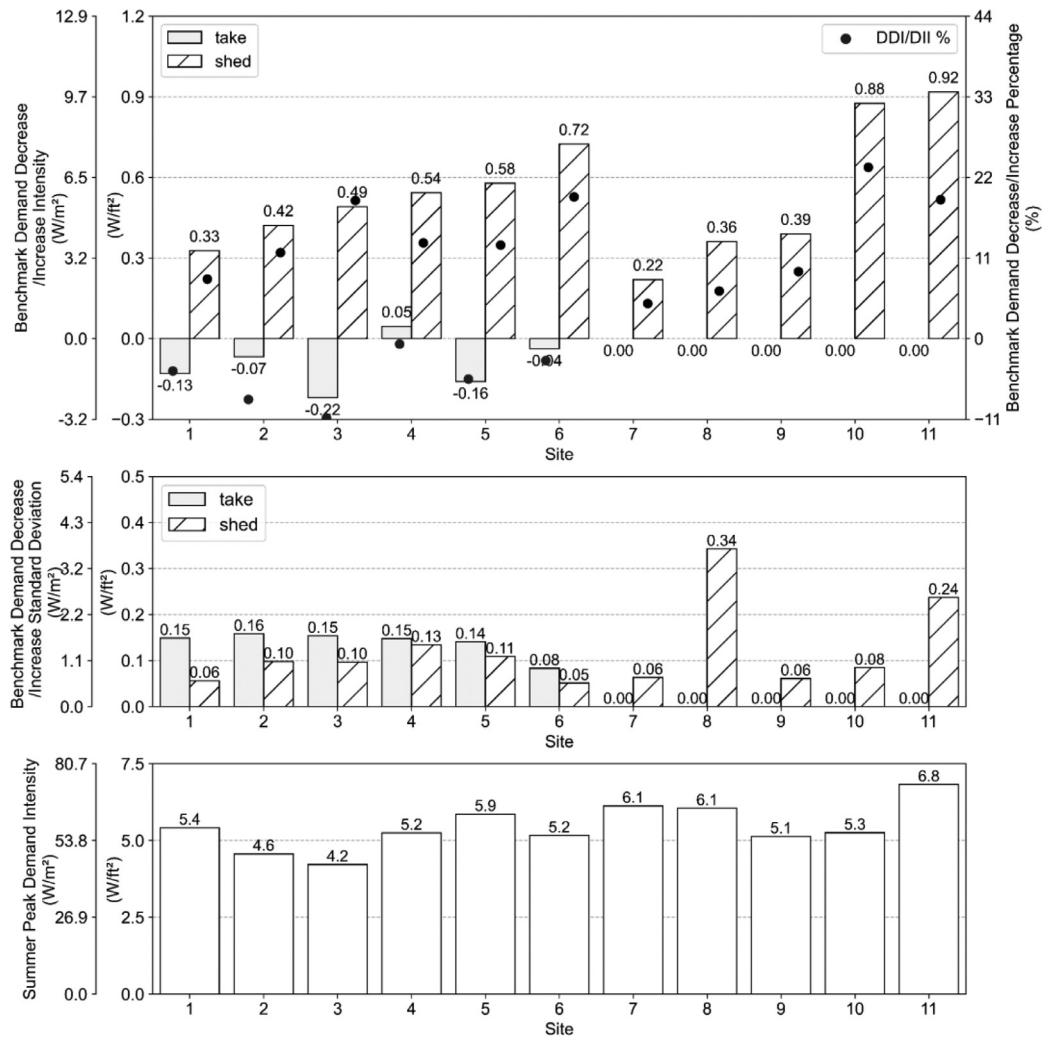


Fig. 9. Benchmarking Metrics for 11 Medium Office Buildings in California That Participated in DR Events (Top: Benchmark DDI, Benchmark DDP, Benchmark DII, Benchmark DIP; Middle: Benchmark DSD, Benchmark DISD; Bottom: SPDI).

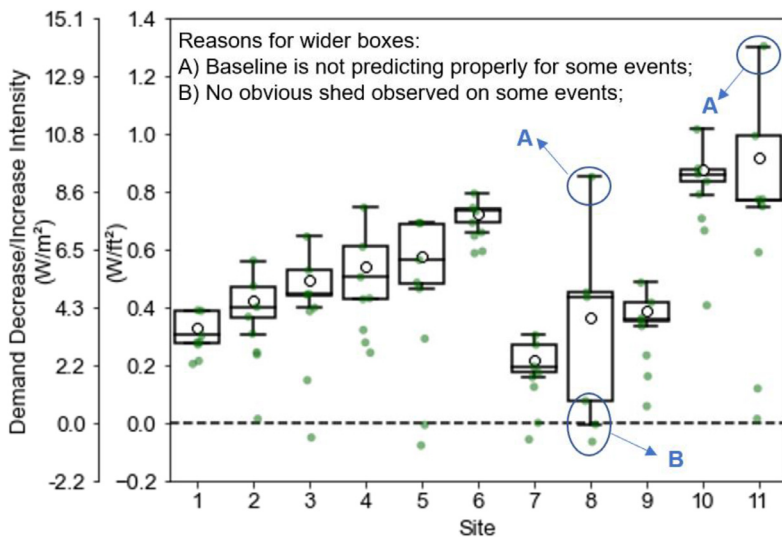


Fig. 10. Boxplot of Single-event DDI Metric (D1) for 11 Medium Office Buildings. (Each dot represents a DR event; the boxes are based on five DR events with the highest D1 values.)

was also observed on the Benchmark DDSD plot in Fig. 9. This could be due to baseline issues and no observed load shed on certain events (as notated in Fig. 10.) For the six buildings that deployed pre-cooling, the Benchmark DDI values are in the range of -0.2 – 0 W/ft^2 (-2 – 0 W/m^2), and Benchmark DIP values are in the range of -10% – 0% . This shows that the Take from pre-cooling is significantly smaller than the Shed, which makes the deployed load shifting strategy energy-efficient. Last but not least, the SPDI are in the range of 4.2 – 6.8 W/ft^2 (45 – 73 W/m^2) and do not display correlation with Benchmark DDI.

4.1.2. Retail buildings example

As mentioned, there is a gap of data showing the DF metrics magnitude, and variation in real building groups of significant size. The retail buildings dataset in this paper presents such an example. The DR event durations and start times are different for these buildings. For the purpose of zero-dimensional benchmarking, a common set of Attribute values were selected:

- **DF strategy:** turning off 50% of the lights and implementing a 3°F (1.7°C) GTA.
- **Event duration:** 2 hours¹⁸
- **Time of day:** 12–6 pm
- **Day of week:** every day
- **Year:** measured in 2021
- **Baseline method:** hourly weather regression
- **Weather condition:** all weather

The procedure of calculating the benchmarking metrics for this building group, shown in Figs. 11 and 12, is similar to the office buildings example, so it is not repeated. After filtering with the above set of Attribute values, 94 of the 121 buildings were included in the zero-dimensional benchmarking metrics results, as shown in Fig. 11, which includes Benchmark DDI (ranked in ascending order), Benchmark DDP, Benchmark DDSD, SPDI, and WPDI. The trend of the Benchmark DDP generally follows the Benchmark DDI; the Benchmark DDSD shows a general descending trend, although it is not consistent from building to building. In terms of load shed magnitude, the Benchmark DDI metric values are in the range of -0.2 – 1.0 W/ft^2 (-2 – 11 W/m^2); the Benchmark DDP metric values are in the range of -10% – 40% . The negative values found in a few buildings are primarily results of no observed shed in the load shape and/or underpredicted baseline load shape. The SPDI and WPDI are in the range of 2.1 – 4.3 W/ft^2 (23 – 46 W/m^2) and 1.4 – 3.2 W/ft^2 (15 – 34 W/m^2), respectively; they do not display correlation with the Benchmark DDI.

Fig. 12 shows the distribution of Single-event D1 values of all non-evening two-hour events in 2021 for each building. The mean value (shown as a dot) of each box matches the Benchmark DDI in Fig. 11. The majority of these buildings have between 0.05 W/ft^2 (0.5 W/m^2) and 0.25 W/ft^2 (2.7 W/m^2) as the range of their top five D1 values (the distance between the top and bottom whiskers), which is also reflected in the low Benchmark DDSD values (in Fig. 11). Some buildings have a much wider range across the top five D1 values, though; they were investigated by reviewing the actual versus baseline daily load shape time series. Four categories of reasons are identified as notated in Fig. 12.

The range of Benchmark DDI values is similar between the two groups of office and retail buildings (0.2 – 0.9 W/ft^2 [2 – 10 W/m^2] versus -0.2 – 1.0 W/ft^2 [-2 – 11 W/m^2]). However, SPDI is significantly higher in the office buildings¹⁹ (4.2 – 6.8 W/ft^2 [45 – 73 W/m^2]) compared to the retail buildings (2.1 – 4.3 W/ft^2 [23 – 46 W/m^2]). Therefore, the relative load shed measured by Benchmark DDP is significantly lower in the office buildings (5% – 25% versus -10% – 40%). While it is not in the scope

of this paper to analyse all influential factors that account for this result due to limited information, it is worth noting that the retail buildings deployed a lighting shed strategy in addition to GTA. This is a major factor in the higher load shed percentage achieved.

4.2. One-Dimensional (1-D) benchmarking

One critical intent of the proposed benchmarking framework design is to enable users to analyse trends and correlations between buildings' DF performance and various influential factors. One-dimensional benchmarking is the simplest form of it, where one of the Attributes is selected as a Dimension to allow observation of any repeated patterns by this dimension building by building. These results can be then further aggregated across buildings to reveal any consistent trends that exist, as illustrated in Fig. 4. This section explores the application of one-dimensional benchmarking to the retail building group for insights; examples include performance year, shed event duration, and time-of-day for the event.

4.2.1. Benchmarking by performance year

The same building's DF performance can change significantly over time due to operational changes, equipment retrofit or maintenance, and other factors. Therefore, it is often useful to examine the trend of a building's DF metrics by year. Fig. 13 shows such an example using eight buildings in New York state with their Benchmark DDI values during 2018 to 2021. A substantial increase of ~ 0.2 – 0.4 W/ft^2 (2 – 4 W/m^2) is observed for all of these buildings for 2020 and 2021.

One primary reason that accounts for such a significant increase in 2020 is that the stores had upgraded their GTA control algorithm for all the sites between the DR seasons of 2019 and 2020. After the control sequence upgrade, the DR participating cooling zones would consistently increase space temperature setpoint by 3°F (1.7°C) during a DR event, whereas in earlier years the temperature adjustment varied significantly by each RTU, based on the old control algorithm. In addition, the selection of RTUs that participate in DR events had also changed at many sites when the control sequence upgrade was implemented. Note that the timing of this upgrade also coincided with the current COVID-19 pandemic. Information is not available to verify the root causes of the metrics results presented, however, the building operations staff believes that the pandemic had a small contribution in the shed increase, given that the HVAC operation was kept mostly the same. It is not in the scope of this paper to fully interpret the complex nuances in the results building by building.

4.2.2. Benchmarking by event duration

These retail buildings in different states participated in DR events of various durations ranging from one to six hours long. Note that these buildings did not curtail lighting during one-hour events out of customer satisfaction considerations; therefore, one-hour events were excluded from this benchmarking example to keep the metrics directly comparable. In Fig. 14, we use 15 buildings in California as an example to investigate potential trends in shed (using the Benchmark DDI metric) in relation to the shed event duration. A consistent declining pattern by event duration was observed in most buildings in this group. Similar patterns were observed in additional states although some states have significant gaps in the event durations (e.g., no 4–5 hour long events). Therefore, the building level metrics were aggregated to state level to identify general trends, as shown in Fig. 15. Such aggregation techniques can be particularly useful for identifying trends that may be difficult to observe building-by-building given such gaps in the data.

Fig. 15 shows Benchmark DDI averaged for all the sites in each state by event duration; an overall average trend line for all sites is also included to evaluate how the implemented strategies work for longer versus shorter events. Overall, a declining trend by event duration was observed, from ~ 0.5 W/ft^2 (5 W/m^2) for the shorter events to ~ 0.2 W/ft^2 (2 W/m^2) for the longer events. A couple factors can contribute to the declining trend in load shed from implementing the GTA strategy as the

¹⁸ Two-hour events are the most common among all the DR events in which these retail buildings participated.

¹⁹ These office buildings have window-to-wall ratios in the range of 40% – 60% , while the retail buildings have few windows.

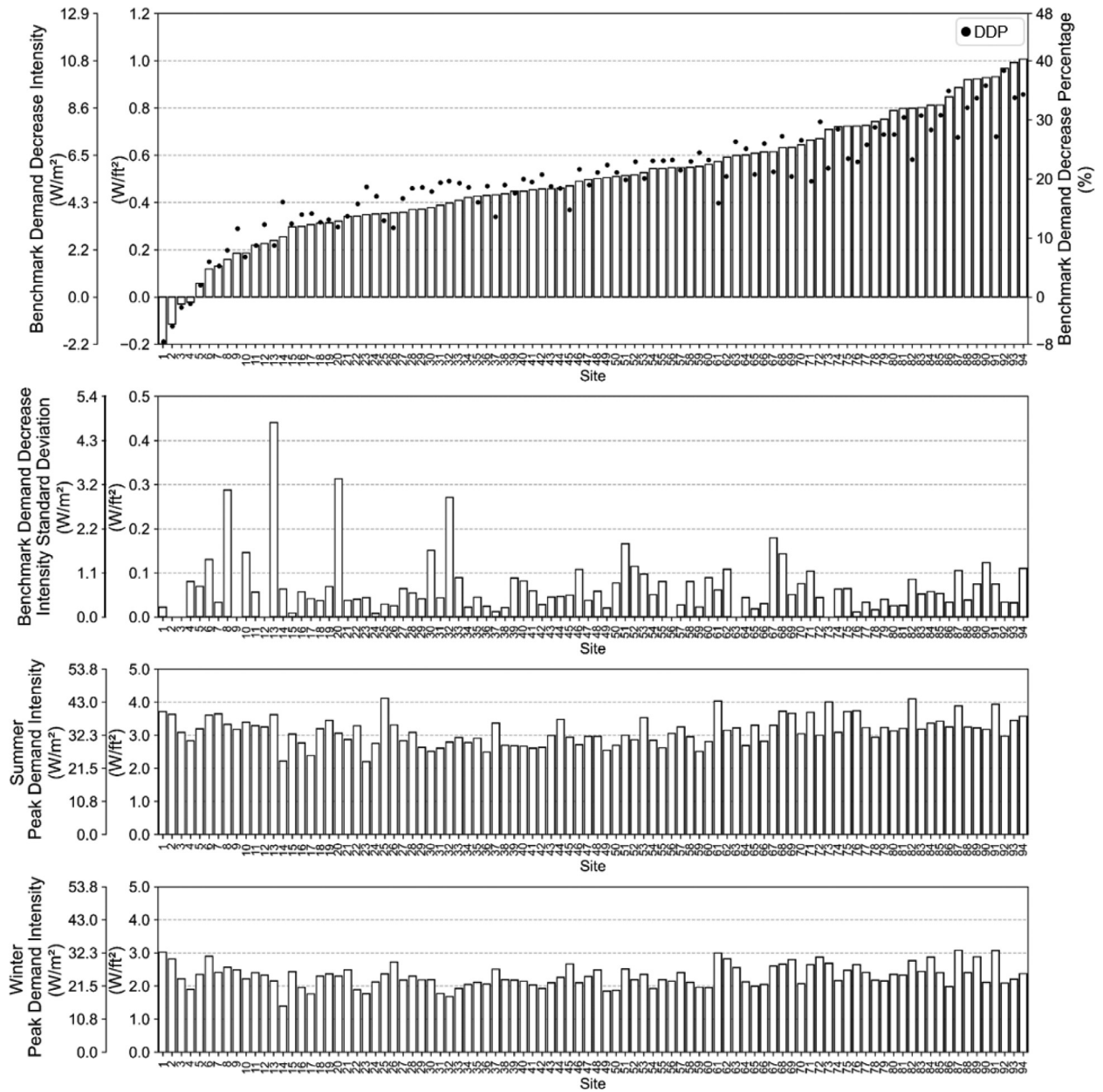


Fig. 11. Benchmarking Metrics for 94 Big-box Retail Buildings That Participated in DR Events, Two-hour Events Using 2021 Data (Top: Benchmark DDI, Benchmark DDP; Second: Benchmark DDSD; Bottom Two: SPDI, WPDI).

event duration increases: (1) HVAC load shed from GTA will reduce significantly once the space temperature reaches the new setpoint (i.e., at the end of “coasting”); (2) as the longer events extend from afternoon into the evening, the HVAC baseline load decreases, and therefore reduces shed.

4.2.3. Benchmarking by time of day

The change of building loads, especially HVAC loads throughout the day, significantly impacts DF metrics. Fig. 16 is an example of one retail building in the group that experienced DR events in both afternoons and evenings. To evaluate the influence of time of day, the two-hour shed duration was kept the same for both afternoon (12–6 pm) and evening (6 pm–midnight) events. The Benchmark DDI is higher for the afternoon events compared to the evening events. The lower cooling load and higher HVAC cooling efficiency during the cooler evening periods may account for this result. It also shows that the variation of load shed across events is smaller in the evening than in the afternoon.

4.3. Two-Dimensional (2-D) benchmarking

In addition to evaluating trends in DF metrics by one Attribute at a time, 2-D benchmarking is another tool available for identifying complex dependencies on two independent Attributes in two dimensions, as illustrated in Fig. 4. Given the nature of it, 2-D benchmarking often requires a significant dataset to be useful and therefore is generally done at an aggregate level. This section explores visualizing Single-event DDI against both shed event duration and daily peak OAT as two dimensions with a subset of the retail buildings in California, as shown in Fig. 17. These buildings have experienced DR events under a wide range of weather conditions, not only on peak summer days. Previous studies have shown that HVAC load shed sometimes has a positive correlation with hotter outdoor temperatures [22, 23, 25]. As shown in Section 4.2.2, the average load shed during an event decreases when event duration increases. We use Fig. 17 to verify if both trends versus weather and event duration can be found in this group of buildings.

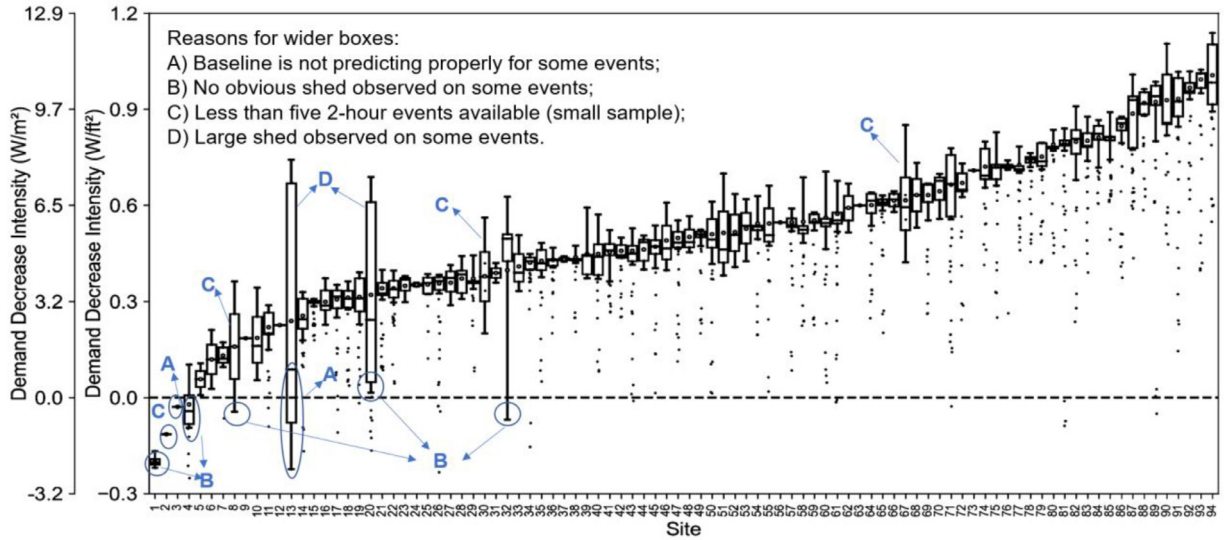


Fig. 12. Boxplot of Single-event DDI Metric (D1) for 94 Big-box Retail Buildings That Participated in DR Events, 2021 Data (each dot represent a DR event; the boxes are based on five DR events with the highest D1 values).

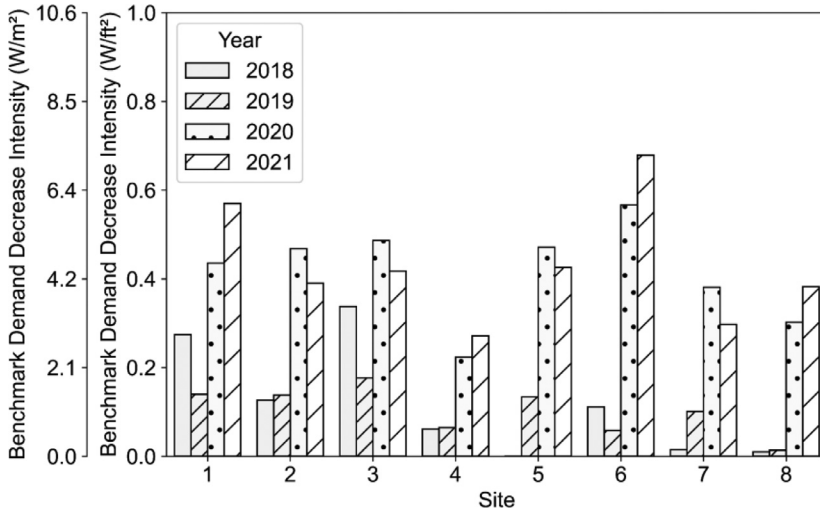


Fig. 13. Benchmark DDI for Two-Hour Events for a Group of Retail Buildings in New York (2018 to 2021).

Given the 1-D benchmarking results in Section 4.2.3, only DR events longer than one-hour happened during the 12–6 pm time frame in 2020 (after the control sequence upgrade) are included in the 2-D plot.

Fig. 17 shows that the highest load shed in the 0.8–1.2 W/ft² (9–13 W/m²) range mostly occurred on the hottest days (~100°F [38 °C] or above) although the load shed on a small fraction of the hot days was significantly lower than 0.6 W/ft² (6 W/m²). In general, the highest values of Single-event DDI in each duration group showed an increasing trend as the event duration got shorter from five hours to two hours. This result implies that the influences from these two Attributes (event duration and weather) are evident when they are reviewed in combination, although there are other factors at play, i.e., the DF metrics in these retail buildings are complex. Although detailed information is not available to fully interpret the results, the building operations staff has shared that factors that may have contributed to the complexity in these results may include HVAC equipment commissioning and maintenance issues, as well as other stochastic factors in the business operations.

5. Discussion

In this paper, a first-of-its-kind DF benchmarking metrics framework is presented and applied to two real building groups as case studies.

The retail building case study was inspired by business values of the building owner's organization (a research partner); in other words, it is a real-world application. This section generalizes applications for building management stakeholders and further explores potential applications and values of this framework for other stakeholder groups. To unlock these potential business values from DF benchmarking, this section also discusses how to make the appropriate data available in terms of contents and format. Once the appropriate dataset has been prepared, it is also important to recognize the significant influence on DF metrics results from the choice of baseline method. Some key considerations are discussed.

5.1. Building Df benchmarking applications

Among the many stakeholders in the ecosystem of building demand flexibility, **property management professionals**, **DF service providers (i.e., aggregators)**, and **utility DF (including DR) programs** are the closest stakeholder groups to benchmarking DF. Property management also consists of various stakeholders who own, operate, or manage a single building or a portfolio of buildings. Each of these stakeholders typically has their key business interests when it comes to DF benchmarking. They are identified in Fig. 18. Building occupants are

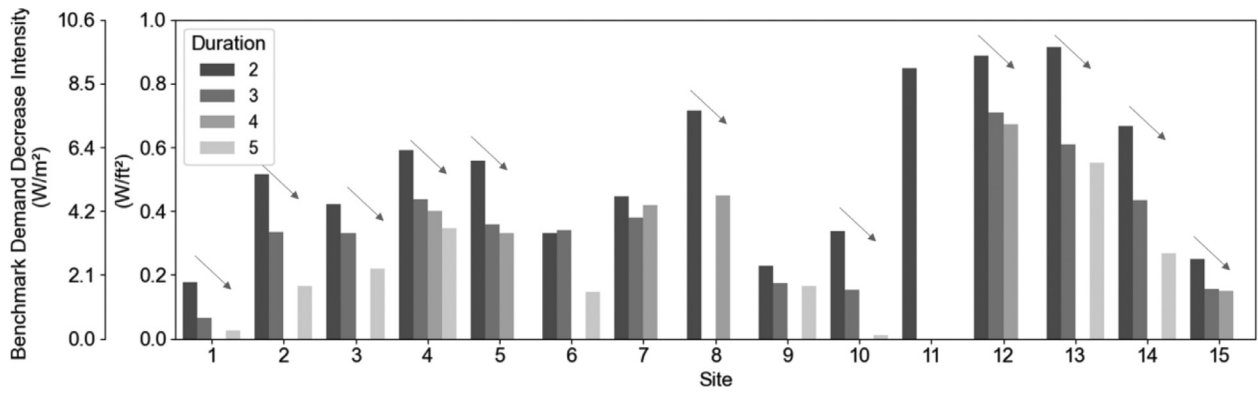


Fig. 14. Benchmark DDI Metric by Shed Event Duration for 15 Retail Buildings in California (2020).

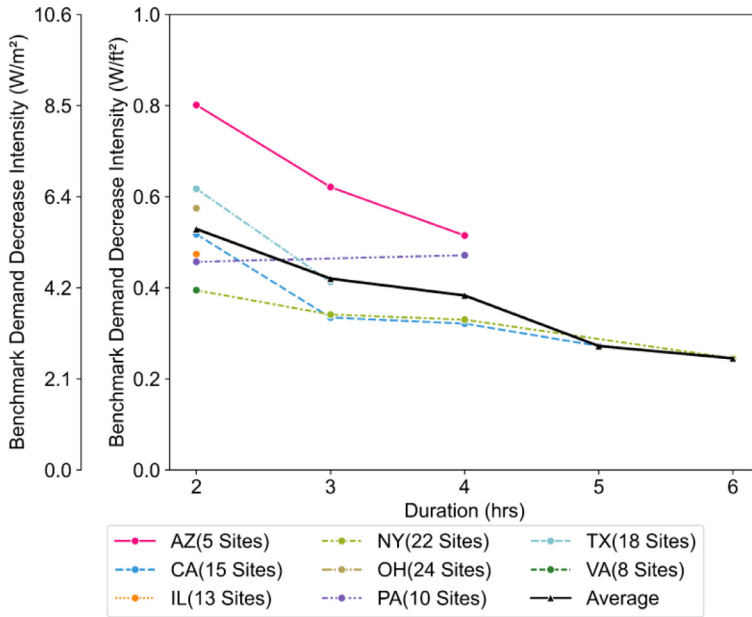


Fig. 15. Benchmark DDI Metric for a Group of 121 Retail Buildings Averaged by State to Show the Trend by Shed Event Duration (2020).

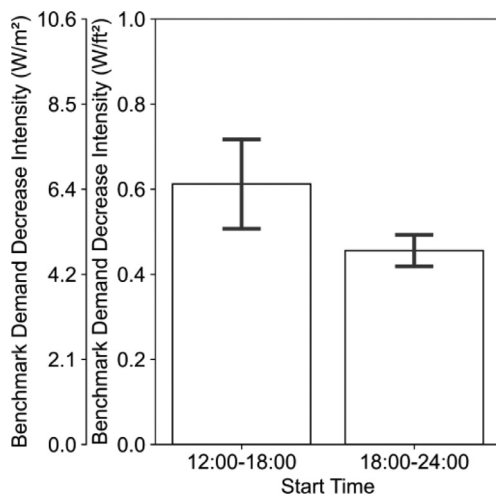


Fig. 16. Benchmark DDI Metric with Standard Deviation for Afternoon versus Evening Two-Hour Events for a Retail Building in California (2021).

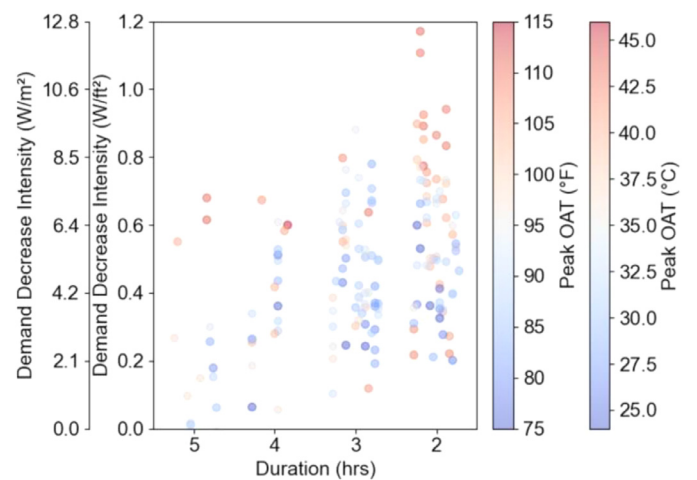


Fig. 17. Two-dimensional Plot for 16 Retail Buildings in California: Single-event DDI Metric by Shed Event Duration with colour Scale Showing Event Day Peak Outside Air Temperature (All During 12–6 pm in 2020).

not directly involved in benchmarking although they do have a role to support achieving building DF performance depending on the deployed strategy - manual strategies sometimes require actions from occupants

(e.g. turn off lights or office equipment) and automatic strategies sometimes allow occupants override - so their buy-in is necessary.

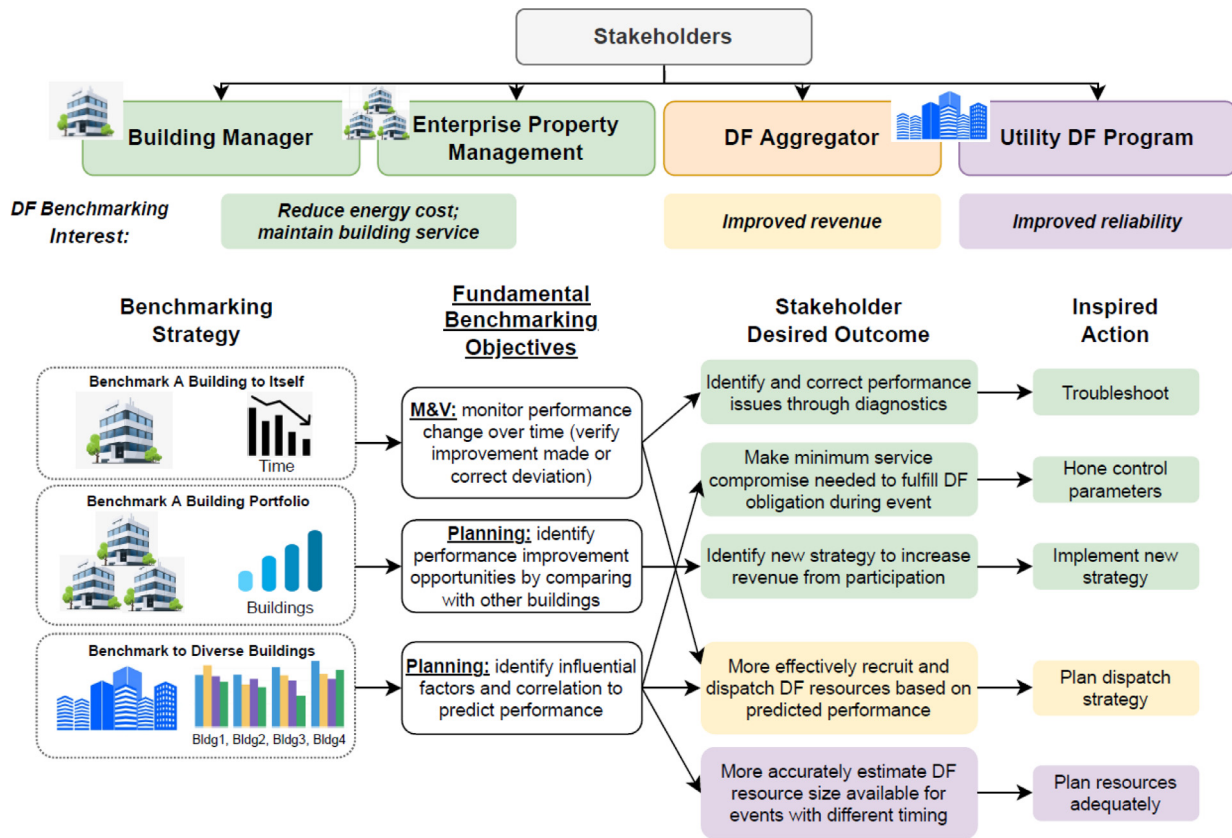


Fig. 18. Potential Applications of DF Benchmarking by Stakeholder Group.

DF benchmarking should be tailored to individual applications depending on the unique business interests, building assets, and data availability. However, there are three fundamental common themes in benchmarking objectives:

- To monitor metrics trend as a function of time:** DF performance deviates over time, so it is important to detect and diagnose such deviations. Another need is to verify performance change when a change or upgrade is made to either equipment or controls, which are expected to impact DF performance. This objective is of interest to both building management professionals and aggregators.
- To identify DF performance improvement opportunities by comparing a building with other buildings:** A fundamental motive of benchmarking is to understand what DF performance can be achieved through what strategies. This objective is of interest to enterprise property management and aggregators.
- To identify influential factors and correlations to predict DF performance:** DF performance is subject to the influence of many factors. It is of interest to property management, aggregators, and utility DF programs to better predict DF performance given a set of conditions (e.g., timing, weather). The business value to each group is discussed below.

Fig. 18 shows how each of the above three common objectives could serve each stakeholder group in achieving their desired outcome from DF benchmarking. The first two objectives also apply to EUI benchmarking; the third objective – **predicting performance** – is unique to DF benchmarking and warrants more discussion here. For property management professionals, it is desirable to achieve greater DF with little compromise to building services. It is common for DF programs to require a commitment level (i.e., minimum kilowatt shed) from the participating buildings. Some programs compensate for higher performance beyond the commitment level to a certain limit; sometimes, programs

may have an under-performance penalty. This means building management can be at loss for financial compensation or building service if their building over-performed or under-performed during an event. Therefore, understanding when higher or lower performance can be expected given a set of event conditions will help building management to tweak their action plan and minimize unnecessary revenue loss and/or building service sacrifice.

For aggregators, there is also a similar motive: to avoid under-performance penalties and uncompensated over-performance. In addition, understanding influential factors will help them prioritize and optimize resource mix in both recruitment and dispatch actions based on those characteristics. This interest from a portfolio standpoint is also relevant to utility DF programs. DF programs often serve utility and grid reliability purposes; therefore, the predictability of DF resources as compared to other resource types (e.g., traditional generation, utility-scale storage) is a critical consideration. A growing understanding of DF predictability and influential factors through benchmarking analysis will help utilities and grid operators more accurately estimate the available DF resource sizes under different conditions. Such understanding would enable them to more effectively use DF resources (amongst other resource types) to address any given grid conditions at hand.

In each unique benchmarking application, a particular objective can be achieved through tailoring the selection of cohort, choice of metrics, applying filters on attribute values, and selection of dimension for visualization, as illustrated in Fig. 3 and Fig. 4.

5.2. Considerations in calculating metrics with real building data

To conduct DF benchmarking using the presented framework, the following data and information is needed for calculating single-event metrics and associated attributes.

Table 9

Summary of Metrics Calculations Supported by Two Different Formats of Building Electricity Consumption Data.

Benchmarking Metrics	Single-event Metrics	Hourly Interval Data for a Year	Kilowatt Shed per DR Event and Monthly Bills
—	Demand Decrease / Increase (D2, I2) (kW)	Yes	Yes - D2 (cannot change baseline)
Benchmark DDI (W/ft ² , W/m ²)	Demand Decrease Intensity (D1) (W/ft ² , W/m ²)	-Yes	Yes (cannot change baseline)
Benchmark DII (W/ft ² , W/m ²)	Demand Increase Intensity (I1) (W/ft ² , W/m ²)	Yes	No
Benchmark DDP, DIP (%)	Demand Decrease / Increase Percentage (D3, I3) (%)	Yes	No
Benchmark DDS , DISD (W/ft ² , W/m ²)	(calculated from D1)	Yes	Yes - DDS; No - DISD
—	Net Building Consumption Change Percentage (24 h) (E1) (%)	Yes	No
Summer/Winter Peak Demand Intensities (SPDI, WPDI) (W/ft ² , W/m ²)	—	Yes	Yes

- One year (or one DR season) of whole-building electricity consumption interval data (hourly or more granular)
- DR event dates and starting and ending times
- Building gross floor area
- Weather parameters (as a single-event attribute) such as daily peak dry-bulb and wet-bulb temperatures; if a weather dependant baseline method is used, then hourly outdoor air temperature may be needed for the same period as interval data.

Building electricity consumption interval data in the U.S. is often measured in 15-minute, 30-minute, or hourly intervals, and all suffice. Sometimes interval data may not be available to utility customers, but the utility provides their DR participating customers with load shed performance results for each event. Such data can also be used to calculate key benchmarking metrics, including the Benchmark DDI and Benchmark DDS. When one year of customer utility bills (containing peak demand in each period) is available, SPDI and WPDI can be calculated as useful reference metrics. However, when such data are used instead of the interval data, the baseline method cannot be altered from what the utility used; other metrics such as Benchmark DDP and load take-related metrics cannot be calculated using this type of data. The benchmarking and single-event metrics that are supported by the above two data formats are summarized in Table 9.

5.3. Baseline method considerations

Unlike calculating metrics from simulation results, when calculating metrics using measured data from real buildings, the baseline method can have significant influence on the metrics results. The authors do not prescribe a baseline method in the metrics definitions out of multiple considerations.

First, the baseline method for demand flexibility is a complex developing research topic. There are at least a few categories of baseline methods found in existing DR programs and research literature, including day-matching, weather-matching, control group based [39], and regression model based methods [40]. Each category has its variations. Currently, there is not a single baseline method that is agreed by the industry to be the most accurate for DF applications. Second, based on the authors' experience, which baseline method is more accurate will depend on a building's specific operations - for example, different buildings' load profiles may have different levels of dependency on weather, occupancy or day-of-the-week and different levels of stochasticity.

Third, data and tool availability is another significant factor in selecting baseline methods. For example, some baseline methods require weather data (may be daily peak temperatures, hourly temperatures or solar radiation parameters); others do not. Some methods may also require a longer period of load profile data before each DR event as the "training dataset" for developing the regression model. Some baseline methods also require a sophisticated computer tool in order to implement it and such a tool may not be publicly available. Furthermore, sometimes the baseline method is dictated by the utility or DR program administrator even though it may not be the most accurate method for a given building. With the above considerations, the appropriate baseline method for calculating these metrics for a given building should be selected on a case-by-case basis.

6. Conclusions and future work

In this paper, the authors have proposed a new framework for benchmarking building DF based on a set of metrics focused on load shedding and load shifting drawn from the literature. The benchmarking framework is intended to primarily benefit planning and post-analysis for grid entities and building stakeholders. The novelty of the framework is a contribution considering that (1) it is designed to address the challenges in benchmarking real buildings and emphasizes graphically presenting benchmarking metrics results for insights; (2) it associates a set of attributes describing the load shed or shift event conditions with the single-event metrics values to enable benchmarking in different dimensions; (3) benchmarking in different dimensions allows users to observe trends and patterns at both the individual building level and at aggregate, which enables stakeholders to identify influential factors and correlations; (4) aggregated dimensional benchmarking analysis makes it possible to leverage more data points for insights by combining metrics of diverse building datasets across geographic areas with different climates and grid needs.

To illustrate how the above framework can be applied to benchmarking real buildings at a significant scale, this paper introduces two case studies of 11 office buildings with a load shifting strategy and 121 big-box retail buildings with a load shedding strategy. This is a main contribution of this paper considering the gap of DF performance data from a large number of buildings in literature. These case studies validated the presented calculation procedures and demonstrated dimensional approach based visualization strategies. The range of DF metric

values from the practical building applications provide valuable insights for stakeholders to evaluate and select DF strategies. In the office building case study, Benchmark DDI for six-hour events was in the range of 0.2–0.9 W/ft² (2–10 W/m²) (Benchmark DDP values of 5%–25%); using pre-cooling may potentially improve the consistency of load shed from GTA; and the combined pre-cooling and GTA load shifting strategy is energy-efficient in these buildings (considering the Benchmark DII values are -0.2–0 W/ft² [-2–0 W/m²]). In the retail building case study, reducing lighting and GTA (without pre-cooling) yielded Benchmark DDI metric values of -0.2–1.0 W/ft² (-2–11 W/m²) (Benchmark DDP values of -10%–40%) in two-hour events. The retail building case study also showed (1) the GTA control algorithm upgrade led to a significant load shed increase, (2) the GTA strategy works better for shorter versus longer events, and (3) the timing of the load shed event and daily peak outdoor temperature are also influential factors.

To generalize and expand upon the two case studies, “zero-dimensional” benchmarking is useful for comparing a number of buildings on a level plane. When used on a portfolio of buildings that implemented similar strategies (as in the case studies), it helps the portfolio management understand how a given strategy worked for each building and provides a screening tool for identifying those buildings that need improvement and diagnosis. Using it on a group of buildings that have implemented different strategies, it can reveal additional opportunities. “One-dimensional” benchmarking provides a tool for identifying trends and patterns in DF metrics with respect to a single variable, and it further allows the use of aggregation to extract insights about influential factors from a large data sample. Benchmarking load shed versus event duration in the retail building case study was such an example. “Two-dimensional” benchmarking enables an empirical approach to understand influences from two variables at the same time and generally requires a large data sample.

The authors have identified various applications for key stakeholder groups considering their business interests. By calculating the benchmarking metrics and visualizing them, we can gain information about the magnitude, predictability (from the variability metrics), and correlations of the load shed or shift that can be expected from implementing a (set of) DF strategy(ies) in a given building. From there, stakeholders can take actions to diagnose, improve, estimate, and plan accordingly. In implementing DF metrics for real buildings, data availability and baseline methods are common challenges; the authors provided discussions as general guidance to these issues. The benchmarking framework and the DF case studies and performance dataset have strong synergy with the on-going efforts about flexible load controls and building-grid integration under the IEA EBC Annex 81: Data-driven smart buildings, and Annex 82: Energy Flexible Buildings Towards Resilient Low Carbon Energy Systems.

In terms of future work, one important area would be to apply the framework to large, diverse real building datasets available to researchers and building practitioners, to accumulate empirical evidence based understanding of commercial building DF. This framework can be implemented in a public benchmarking database with a user interface to enable various real-world applications. It is a next step to expand this framework by providing a methodology to accommodate buildings with DERs such as solar photovoltaics, battery storage, and electric vehicles, and experimenting the methodology with a group of real buildings. Another valuable future work would be to link the DF metrics to carbon metrics to inform building decarbonization applications as carbon intensity of grid electricity varies with time, day and season. From the building manager’s standpoint, although meeting building services and occupant comfort requirements are currently treated as a default criteria in the framework, it could be expanded in the future to address concerns around potential negative impact on occupant comfort and productivity when certain DF strategies alter the indoor environment or trade off with building services. It is known to be a challenging area due to data limitations and lack of widely adopted standards across geographical areas.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jingjing Liu: Conceptualization, Methodology, Visualization, Writing – original draft. **Rongxin Yin:** Methodology, Data curation, Resources. **Lili Yu:** Software, Visualization. **Mary Ann Piette:** Conceptualization, Methodology, Writing – review & editing. **Marco Pritoni:** Conceptualization, Writing – review & editing, Funding acquisition. **Armando Casillas:** Resources, Writing – review & editing. **Jiarong Xie:** Data curation. **Tianzhen Hong:** Writing – review & editing. **Monica Neukomm:** Methodology. **Peter Schwartz:** Funding acquisition.

Acknowledgements

This work was supported by the Assistant Secretary for Energy Efficiency and Renewable Energy, Building Technologies Office, of the U.S. Department of Energy under Contract No. DE-AC02–05CH11231. The authors want to thank Ms. Monica Neukomm and Mr. Amir Roth for funding this research work. We also wish to acknowledge the owner organization of the retail buildings included in this study for providing valuable DR performance data and sharing their experiences and insights to support our DF benchmarking research.

References

- [1] EBC. 2014. Annex 67. <https://www.annex67.org/>
- [2] IEA EBC. 2022. Annex 82. <https://annex82.iea-ebc.org/>
- [3] Jensen SØ, Pomianowska AM, Lollini R, Pasut W, Knotzer A, Engelmann P, Stafford A, Reyniers G. IEA EBC annex 67 energy flexible buildings. *Energy Build* 2017;155:25–34 Nov..
- [4] International Energy Agency (IEA). (2019). *Summary report: energy in buildings and communities Annex 67 energy flexible buildings*. <https://annex67.org/media/1920/summary-report-annex-67.pdf>.
- [5] IEA EBC. 2022. Annex 81. <https://annex81.iea-ebc.org/>
- [6] Li H, Wang Z, Hong T, Piette MA. Energy flexibility of residential buildings: a systematic review of characterization and quantification methods and applications. *Adv Appl Energy* 2021;3:100054 Aug.. doi:10.1016/j.adapen.2021.100054.
- [7] Neukomm, M., Nubbe, V., & Fares, R. (2019). *Grid-interactive efficient buildings technical report series*. (December). Retrieved on April 8, 2022 from <https://www1.eere.energy.gov/buildings/pdfs/75470.pdf>.
- [8] Gerke B, Zhang C, Satchwell A, Murthy S, Piette M, Present E, Wilson E, Speake A, Adhikari R. Modeling the interaction between energy efficiency and demand response on regional grid scales. 2020 ACEEE Summer Study on Energy Efficiency in Buildings (Virtual Conference); 2020 <https://www.nrel.gov/docs/fy20osti/77423.pdf>.
- [9] Langevin J, Harris CB, Satri-Meloy A, Chandra-Putra H, Speake A, Present E, Adhikari R, Wilson EJ, Satchwell AJ. US building energy efficiency and flexibility as an electric grid resource. *Joule* 2021;5:1–27. doi:10.1016/j.joule.2021.06.002.
- [10] Gerke, B., Gallo, G., Smith, S., Liu, J., Alstone, P., Raghavan, S., Schwartz, P., Piette, M.A., Yin, R., & Stensson, S. 2020. *The California demand response potential study, Phase 3: final report on the shift resource through 2030*. <https://doi.org/10.20357/B7MS40>.
- [11] ENERGY STAR. No date. Benchmark your building using ENERGY STAR® Portfolio Manager® <https://www.energystar.gov/buildings/benchmark>
- [12] Mathew PA, Walter T. Is the bpd nationally representative? A comparison of the building performance database to the commercial buildings energy consumption survey. Lawrence Berkeley National Laboratory; 2019. 2019LBNL-20001198.
- [13] Liu J, Yin R, Pritoni M, Piette MA, Neukomm M. Developing and evaluating metrics for demand flexibility in buildings: comparing simulations and field data. 2020 ACEEE Summer Study on Energy Efficiency in Buildings (Virtual Conference); 2020. doi:10.20357/B7WW34.
- [14] Miller A, Carbonnier K. New metrics for evaluating building-grid integration. 2020 ACEEE Summer Study on Energy Efficiency in Buildings (Virtual Conference); 2020.
- [15] Miller, A., Carbonnier, K. & Egolf, M. (2021). *Year three progress and outcomes*. <https://newbuildings.org/wp-content/uploads/2022/01/GridOptimal-Year-3-Outcomes-Summary-December-2021.pdf>.
- [16] Airò Farulla G, Tumminia G, Sergi F, Aloisio D, Cellura M, Antonucci V, Ferraro M. A review of key performance indicators for building flexibility quantification to support the clean energy transition. *Energies* 2021;14:5676. doi:10.3390/en14185676.
- [17] Zhang L, Good N, Mancarella P. Building-to-grid flexibility: modeling and assessment metrics for residential demand response from heat pump aggregations. *Appl Energy* 2019;709–23 233–234 (January). doi:10.1016/j.apenergy.2018.10.058.

- [18] Gerke BF, Zhang C, Murphy S, Satchwell AJ, Present E, Horsey H, Wilson E, Parker A, Speake A, Adhikari R, Piette MA. Load-driven interactions between energy efficiency and demand response on regional grid scales. *Adv Appl Energy* 2022;6:100092. doi:10.1016/j.adapen.2022.100092.
- [19] Aduda KO, Labeodan T, Zeiler W. Towards critical performance considerations for using office buildings as a power flexibility resource - a survey. *Energy Build* 2018;159:164–78. doi:10.1016/j.enbuild.2017.10.096.
- [20] Li R, You S. Exploring potential of energy flexibility in buildings for energy system services. *CSEE J Power Energy Syst* 2018;4(4):434–43. doi:10.17775/CSEE-JPES.2018.00360.
- [21] Johra, H., Marszal-pomianowska, A., Ellingsgaard, J.R., & Liu, M. (2019). *Building energy flexibility : a sensitivity analysis and key performance indicator comparison*. <https://doi.org/10.1088/1742-6596/1343/1/012064>.
- [22] Luo N, et al. Quantifying the effect of multiple load flexibility strategies on commercial building electricity demand and services via surrogate modeling. *Appl Energy* 2022;309. doi:10.1016/j.apenergy.2021.118372.
- [23] Cai J, Braun JE. Assessments of demand response potential in small commercial buildings across the United States. *Sci Technol Built Environ* 2019;25(10):1437–55. doi:10.1080/23744731.2019.1629245.
- [24] Keskar A, Lei S, Webb T, Nagy S, Hiskens IA, Mathieu JL, Johnson JX. Assessing the performance of global thermostat adjustment in commercial buildings for load shifting demand response. *Environ Res: Infrastruct Sustainab* 2022;2. doi:10.1088/2634-4505/ac51c5.
- [25] Piette MA, Watson D, Motegi N, Kiliccote S, Xu P. Automated critical peak pricing field tests: program description and results. Lawrence Berkeley National Laboratory; 2006. LBNL-59351.
- [26] Kiliccote S, Piette MA, Mathieu J, Parrish K. Findings from seven years of field performance data for automated demand response in commercial buildings. 2010 ACEEE Summer Study on Energy Efficiency in Buildings; 2010.
- [27] Yin R, Xu P, Piette MA, Kiliccote S. Study on Auto-DR and pre-cooling of commercial buildings with thermal mass in California. *Energy Build*. 2010;42(7):967–75. doi:10.1016/j.enbuild.2010.01.008.
- [28] Yin R, Kara EC, Li Y, DeForest N, Wang K, Yong T, Stadler M. Quantifying flexibility of commercial and residential loads for demand response using setpoint changes. *Appl Energy* 2016;177:149–64. doi:10.1016/j.apenergy.2016.05.090.
- [29] Eng MAL, Lavigne Jr, K, Poulin A. Demand response in commercial buildings: a cold climate field study. *ASHRAE Transactions*, 121, 1U; 2015. Retrieved from <https://www.proquest.com/scholarly-journals/>.
- [30] Fournier M, Leduc M-A, Nadrault G. Winter demand response using baseboard heaters: achieving substantial demand reduction without sacrificing comfort. *Proc. ACEEE Summer Study Energy Efficiency. Build.* (Pacific Grove, CA) 2016;828:1–12 Csa 2013.
- [31] Hirsch, A., Clark, J., Deru, M., Trenbath, K., Doebber, I., & Studer, D. (2015). *Pilot Testing of Commercial Refrigeration-Based Demand Response*. Retrieved from www.nrel.gov/publications.
- [32] Newsham G, Birt B. Demand-responsive lighting – a field study. *LEUKOS* 2010;6(3):203–26. doi:10.1582/LEUKOS.2010.06.03002.
- [33] ENERGY STAR. Climate and weather. Portfolio manager technical reference. https://www.energystar.gov/sites/default/files/tools/Climate_and_Weather_2020_508.pdf.
- [34] Kim J, Yin R, Kiliccote S. Automated price and demand response demonstration for large customers in New York City using OpenADR. 2013 International Conference for Enhanced Building Operations Montreal, Quebec; 2013. October 8–10, 2013.
- [35] Mathieu JL, Price P, Kiliccote S, Piette MA. Quantifying changes in building electricity use, with application to demand response. *IEEE Trans Smart Grid* 2011;2:507–18.
- [36] ASHRAE ASHRAE guideline 14-2014 – measurement of energy, demand and water savings. Atlanta, GA: American Society of Heating, Refrigeration and Air Conditioning Engineers; 2014.
- [37] Liu J, Yu L, Yin R, Piette MA, Pritoni M, Casillas A, Neukomm M. Benchmarking demand flexibility in commercial buildings and flattening the duck – addressing baseline and commissioning challenges. 2022 ACEEE Summer Study on Energy Efficiency in Buildings; 2022.
- [38] Yin R, Xu P, Kiliccote S. Auto-DR and pre-cooling of buildings at tri-city corporate center. Lawrence Berkeley National Laboratory; 2008. LBNL-3348E.
- [39] Bode J, Ciccone A. California iso baseline accuracy assessment (November 2017). CAISO Baseline Accuracy Working Group; 2017. (Retrieved on March 23, 2022 from <https://www.caiso.com/Documents/CaliforniaISOBaselineAccuracyAssessmentNexant.pdf>).
- [40] Granderson J, Sharma M, Crowe E, Jump D, Fernandes S, Touzani S, Johnson D. Assessment of model-based peak electric consumption prediction for commercial buildings. *Energy Build* 2021;245:111031 ISSN 0378-7788. doi:10.1016/j.enbuild.2021.111031.